Communications

Previous Year Questions • 1987–2026

GATE ELECTRONICS AND COMMUNICATION ENGINEERING

(EC)

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in Communications (EC), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by unit and by year, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

○○○ ●○○ Easy ●●○ Medium ●●● Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

How to Read the Unit Snapshot

Every unit opens with a **Unit Snapshot** box summarizing that unit's real question data:

Questions	the real total for this unit.
MCQ, MSQ, NAT	that same total, split by question type.
1M, 2M	how many of those questions carry 1 or 2 marks.
Descriptive	questions worth more than 2 marks, or whose marks value isn't recorded.
Avg Weightage	the average marks this unit has contributed per real GATE sitting, not a raw year-by-year total — a year GATE ran multiple real sessions (a genuine forenoon/afternoon split, not the same paper's Set A/B/C/D booklet variants) is divided across those sessions first, so a heavily multi-session year cannot silently outweigh a single-session one. The ~ marks it as a rounded figure, not an exact one.
Range	the lowest and highest that same per-sitting value has been, with the year each occurred.

Avg Weightage and Range are computed from **2010 onward** only: GATE dropped from a 150-mark to a 100-mark paper in 2009, but that year itself was a one-off transition (60 questions, not the 65-question format that became standard from 2010 on) — so 2010 is the first year fully comparable to modern papers, and marks-based figures from earlier years are left out of these two figures specifically (every other count on this page still includes every year). They also depend on knowing which GATE years ran more than one real sitting for a given branch — our own research into this is thorough for some branches and still a work in progress for others. Treat both figures as a **beta**, best-effort estimate: directionally useful, not exact.

MSQ questions specifically weren't introduced until 2021 — an older-year-heavy unit showing **MSQ: 0** reflects that real exam history, not a data gap.

Worked example

Unit Snapshot*

Stream: EC	**Questions:** 45	**MCQ:** 30	**MSQ:** 3	**NAT:** 12
1M: 20	**2M:** 22	**Descriptive:** 3	**Avg Weightage** ~4M	**Range:** 2M(2015)–6M(2021)

45 real questions total; on average this unit has been worth about 4 marks per real GATE sitting since 2010, ranging from as low as 2 (in 2015) to as high as 6 (in 2021).

A quick way to use this: a high **Avg Weightage** points to a consistently high-yield topic worth prioritizing; a wide **Range** usually means the topic's importance has shifted over the years, so it's worth checking *which* years drove that before deciding how much weight to give it today.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

Contributors

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UNIT

I

Random Processes

Topics

1. Autocorrelation and Power Spectral Density
2. Properties of White Noise
3. Filtering of Random Signals Through LTI Systems

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Unit Snapshot*

Stream: EC	**Questions:** 69	**MCQ:** 50	**MSQ:** 1	**NAT:** 18
1M: 23	**2M:** 46	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2021)–8M(2013)

**See preface for how Avg Weightage and Range are calculated.*

Q1.26.1 MCQ 2026 2M

Let X, N, Y and Z be random variables. The variables X and N are independent of each other. X is uniformly distributed between -1 and 1; N follows the Normal distribution with zero mean and unity variance.
Y and Z are defined as,

$$Y = X + N \quad \text{and} \quad Z = X^2 + N.$$

Which of the following pairs represents the values of correlation between X and Y and that between X and Z?

A. $\frac{1}{3}$ and 0 **B.** $\frac{1}{3}$ and $\frac{1}{9}$
C. $\frac{1}{3}$ and $\frac{1}{3}$ **D.** 1 and 0

Q1.25.1 MCQ 2025 2M

Consider a real-valued random process

$$f(t) = \sum_{n=1}^{N} a_n \, p(t - nT),$$

where $T > 0$ and N is a positive integer. Here, $p(t) = 1$ for $t \in [0, 0.5T]$ and 0 otherwise. The coefficients a_n are pairwise independent, zero-mean unit-variance random variables.

Read the following statements about the random process and choose the correct option.

(i) The mean of the process $f(t)$ is independent of time t.

(ii) The autocorrelation function $E[f(t)f(t+\tau)]$ is independent of time t for all τ.

(Here, $E[\cdot]$ is the expectation operation.)

A. (i) is TRUE and (ii) is FALSE

B. Both (i) and (ii) are TRUE

C. Both (i) and (ii) are FALSE

D. (i) is FALSE and (ii) is TRUE

Q1.24.1 MCQ 2024 1M

A white Gaussian noise $w(t)$ with zero mean and power spectral density $\frac{N_0}{2}$, when applied to a first-order RC low pass filter produces an output $n(t)$. At a particular time $t = t_k$, the variance of the random variable $n(t_k)$ is _____.

A. $\frac{N_0}{4RC}$ **B.** $\frac{N_0}{2RC}$
C. $\frac{N_0}{RC}$ **D.** $\frac{2N_0}{RC}$

Q1.24.2 NAT 2024 1M

Suppose X and Y are independent and identically distributed random variables that are distributed uniformly in the interval $[0, 1]$. The probability that $X \geq Y$ is _____.

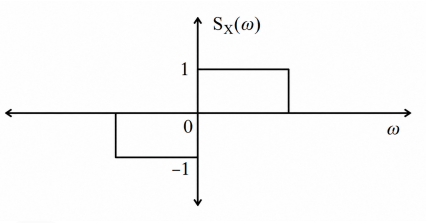
Q1.23.1 MSQ 2023 1M

For a real signal, which of the following is/are valid power spectral density/densities?

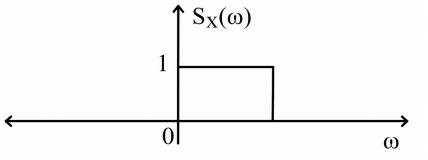
☐ **A.** $S_X(\omega) = \frac{2}{9 + \omega^2}$

☐ **B.** $S_X(\omega) = e^{-\omega^2}\cos^2\omega$

☐ **C.**



☐ **D.**



Q1.23.2 MCQ 2023 2M

A random variable X, distributed normally as $N(0, 1)$, undergoes the transformation $Y = h(X)$, given in the figure. The form of the probability density function of Y is
(In the options given below, a, b, c are non-zero constants and $g(y)$ is piece-wise continuous function)

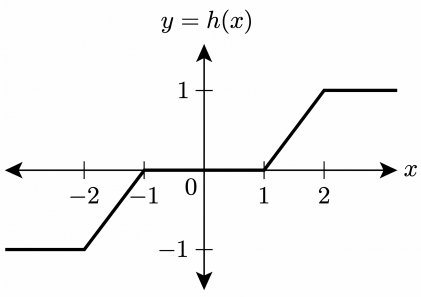


Fig. Q1.23.2

A. $a\delta(y-1) + b\delta(y+1) + g(y)$

B. $a\delta(y+1) + b\delta(y) + c\delta(y-1) + g(y)$

C. $a\delta(y+2) + b\delta(y) + c\delta(y-2) + g(y)$

D. $a\delta(y+2) + b\delta(y-2) + g(y)$

Q1.23.3 NAT 2023 2M

Let $X(t)$ be a white Gaussian noise with power spectral density $\frac{1}{2}$ W/Hz. If $X(t)$ is input to an LTI system with impulse response $e^{-t}u(t)$. The average power of the system output is _____ W *(rounded off to two decimal places).*

Q1.22.1 NAT 2022 2M

Consider a real valued source whose samples are independent and identically distributed random variables with the probability density function, $f(x)$, as shown in the figure.

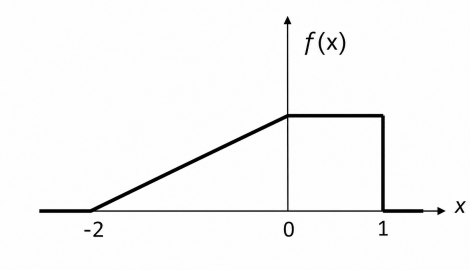


Fig. Q1.22.1

Consider a 1 bit quantizer that maps positive samples to value α and others to value β. If α^* and β^* are the respective choices for α and β that minimize the mean square quantization error, then $(\alpha^* - \beta^*)$ = _____ *(rounded off to two decimal places).*

Q1.21.1 NAT 2021 1M

The autocorrelation function $R_X(\tau)$ of a wide-sense stationary random process $X(t)$ is shown in the figure.

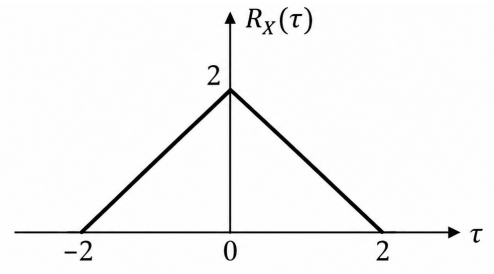


Fig. Q1.21.1

The average power of $X(t)$ is _____

Q1.20.1 NAT 2020 1M

The random variable

$$Y = \int_{-\infty}^{\infty} W(t)\phi(t)\,dt$$

where

$$\phi(t) = \begin{cases} 1, & 5 \le t \le 7 \\ 0, & \text{otherwise} \end{cases}$$

and $W(t)$ is a real white Gaussian noise process with two-sided power spectral density $S_W(f) = 3$ W/Hz, for all f. The variance of Y is _____.

Q1.19.1 NAT 2019 2M

Let a random process $Y(t)$ be described as

$$Y(t) = h(t) * X(t) + Z(t),$$

where $X(t)$ is a white noise process with power spectral density $S_X(f) = 5$ W/Hz. The filter $h(t)$ has a magnitude response given by

$$|H(f)| = 0.5 \quad \text{for } -5 \le f \le 5,$$

and zero elsewhere. $Z(t)$ is a stationary random process, uncorrelated with $X(t)$, with power spectral density as shown in the figure. The power in $Y(t)$, in watts, is equal to _____ W (rounded off to two decimal places).

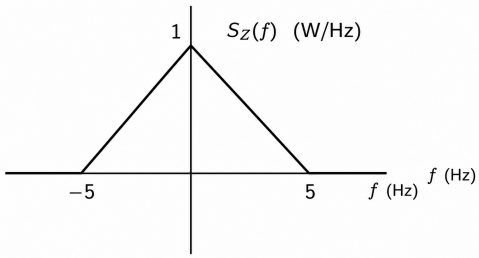


Fig. Q1.19.1

Q1.19.2 NAT 2019 1M

If X and Y are random variables such that

$$E[2X + Y] = 0$$

and

$$E[X + 2Y] = 33,$$

then

$$E[X] + E[Y] = _____.$$

Q1.19.3 NAT 2019 1M

Let Z be an exponential random variable with mean 1. That is, the cumulative distribution function of Z is given by

$$F_Z(x) = \begin{cases} 1 - e^{-x}, & \text{if } x \geq 0 \\ 0, & \text{if } x < 0 \end{cases}$$

Then

$$\Pr(Z > 2 \mid Z > 1),$$

rounded off to two decimal places, is equal to _____.

Q1.18.1 MCQ 2018 2M

Consider a white Gaussian noise process $N(t)$ with two-sided power spectral density

$$S_N(f) = 0.5 \text{ W/Hz}$$

as input to a filter with impulse response

$$0.5e^{-t^2/2}$$

(where t is in seconds) resulting in output $Y(t)$. The power in $Y(t)$ in watts is

A. 0.11 **B.** 0.22
C. 0.33 **D.** 0.44

Q1.18.2 NAT 2018 1M

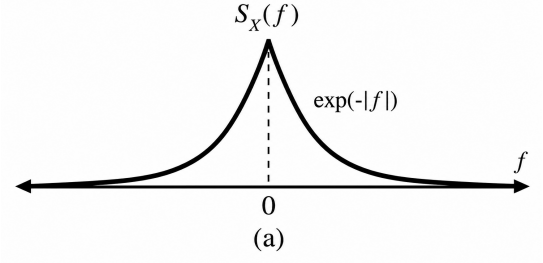
Let X_1, X_2, X_3 and X_4 be independent normal random variables with zero mean and unit variance. The probability that X_4 is the smallest among the four is _____.

Q1.17.1 MCQ 2017 2M

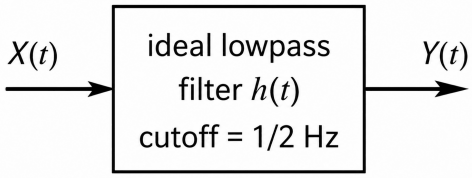
Let $X(t)$ be a wide sense stationary random process with the power spectral density $S_X(f)$ as shown in Figure (a), where f is in Hertz (Hz). The random process $X(t)$ is input to an ideal lowpass filter with the frequency response

$$H(f) = \begin{cases} 1, & |f| \leq \frac{1}{2} \text{ Hz} \\ 0, & |f| > \frac{1}{2} \text{ Hz} \end{cases}$$

as shown in Figure (b). The output of the lowpass filter is $Y(t)$.



(a)



(b)

Let E be the expectation operator and consider the following statements:

1. $E(X(t)) = E(Y(t))$
2. $E(X^2(t)) = E(Y^2(t))$
3. $E(Y^2(t)) = 2$

Select the correct option:

A. only I is true **B.** only II and III are true
C. only I and II are true **D.** only I and III are true

Q1.17.2 NAT 2017 1M

Consider the random process

$$X(t) = U + Vt,$$

where U is a zero-mean Gaussian random variable and V is a random variable uniformly distributed between 0 and 2. Assume that U and V are statistically independent. The mean value of the random process at $t = 2$ is _____.

Q1.16.1 NAT 2016 2M

An information source generates a binary sequence $\{a_n\}$. a_n can take one of the two possible values -1 and $+1$ with equal probability and are statistically independent and identically distributed. This sequence is precoded to obtain another sequence $\{\beta_n\}$, as

$$\beta_n = a_n + k\, a_{n-3}.$$

The sequence $\{\beta_n\}$ is used to modulate a pulse $g(t)$ to generate the baseband signal

$$X(t) = \sum_{n=-\infty}^{\infty} \beta_n g(t - nT),$$

where

$$g(t) = \begin{cases} 1, & 0 \leq t \leq T \\ 0, & \text{otherwise.} \end{cases}$$

If there is a null at

$$f = \frac{1}{3T}$$

in the power spectral density of $X(t)$, then k is _____.

Q1.16.2 NAT 2016 2M

Consider a random process

$$X(t) = 3V(t) - 8,$$

where $V(t)$ is a zero mean stationary random process with autocorrelation

$$R_V(\tau) = 4e^{-5|\tau|}.$$

The power in $X(t)$ is _____.

Q1.16.3 MCQ 2016 2M

A wide sense stationary random process $X(t)$ passes through the LTI system shown in the figure. If the autocorrelation function of $X(t)$ is $R_X(\tau)$, then the autocorrelation function $R_Y(\tau)$ of the output $Y(t)$ is equal to

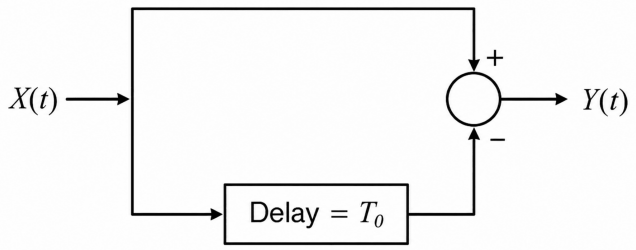


Fig. Q1.16.3

A. $2R_X(\tau) + R_X(\tau - T_0) + R_X(\tau + T_0)$

B. $2R_X(\tau) - R_X(\tau - T_0) - R_X(\tau + T_0)$

C. $2R_X(\tau) + 2R_X(\tau - 2T_0)$

D. $2R_X(\tau) - 2R_X(\tau - 2T_0)$

Q1.15.1 MCQ 2015 2M

A zero mean white Gaussian noise having power spectral density $\frac{N_0}{2}$ is passed through an LTI filter whose impulse response $h(t)$ is shown in the figure. The variance of the filtered noise at $t = 4$ is

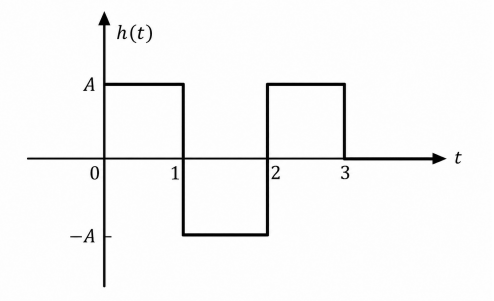


Fig. Q1.15.1

A. $\frac{3}{2}A^2N_0$ **B.** $\frac{3}{4}A^2N_0$

C. A^2N_0 **D.** $\frac{1}{2}A^2N_0$

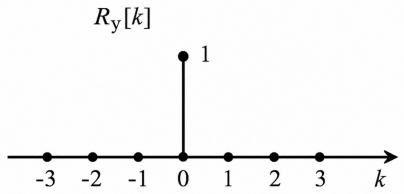
Q1.15.2 MCQ 2015 2M

$\{X_n\}_{n=-\infty}^{\infty}$ is an independent and identically distributed (i.i.d.) random process with X_n equally likely to be +1 or −1. $\{Y_n\}_{n=-\infty}^{\infty}$ is another random process obtained as

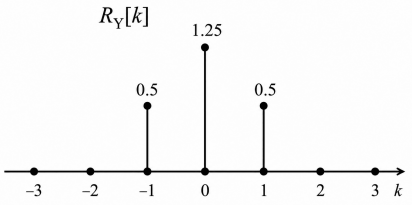
$$Y_n = X_n + 0.5X_{n-1}.$$

The autocorrelation function of $\{Y_n\}_{n=-\infty}^{\infty}$, denoted by $R_Y[k]$, is

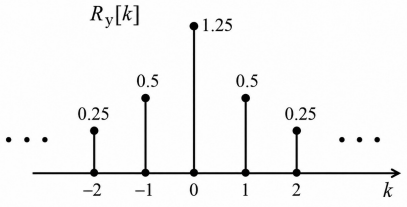
A.



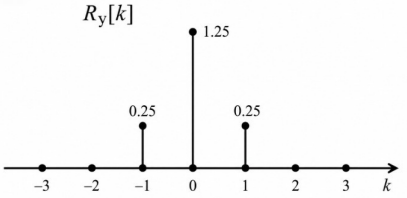
B.



C.



D.



Q1.15.3 MCQ 2015 2M

Let $X \in \{0, 1\}$ and $Y \in \{0, 1\}$ be two independent binary random variables.
If $P(X = 0) = p$ and $P(Y = 0) = q$, then $P(X + Y \geq 1)$ is equal to

A. $pq + (1 - p)(1 - q)$ **B.** pq
C. $p(1 - q)$ **D.** $1 - pq$

Q1.15.4 NAT 2015 2M

A random binary wave $y(t)$ is given by

$$y(t) = \sum_{n=-\infty}^{\infty} X_n \, p(t - nT - \phi)$$

where

$$p(t) = u(t) - u(t - T),$$

$u(t)$ is the unit step function and ϕ is an independent random variable with uniform distribution in $[0, T]$. The sequence $\{X_n\}$ consists of independent and identically distributed binary valued random variables with $P\{X_n = +1\} = P\{X_n = -1\} = 0.5$ for each n.The value of the autocorrelation

$$R_{yy}\left(\frac{3T}{4}\right) \triangleq E\left[y(t)\, y\left(t - \frac{3T}{4}\right)\right]$$

equals _____.

Q1.14.1 MCQ 2014 2M

Consider a random process

$$X(t) = \sqrt{2}\sin(2\pi t + \varphi),$$

where the random phase φ is uniformly distributed in the interval $[0, 2\pi]$. The auto-correlation $E[X(t_1)X(t_2)]$ is

A. $\cos(2\pi(t_1 + t_2))$ **B.** $\sin(2\pi(t_1 - t_2))$
C. $\sin(2\pi(t_1 + t_2))$ **D.** $\cos(2\pi(t_1 - t_2))$

Q1.14.2 NAT 2014 2M

Let $Q(\sqrt{\gamma})$ be the BER of a BPSK system over an AWGN channel with two-sided noise power spectral density $N_0/2$. The parameter γ is a function of bit energy and noise power spectral density.
A system with two independent and identical AWGN channels with noise power spectral density $N_0/2$ is shown in the figure. The BPSK demodulator receives the sum of outputs of both the channels.

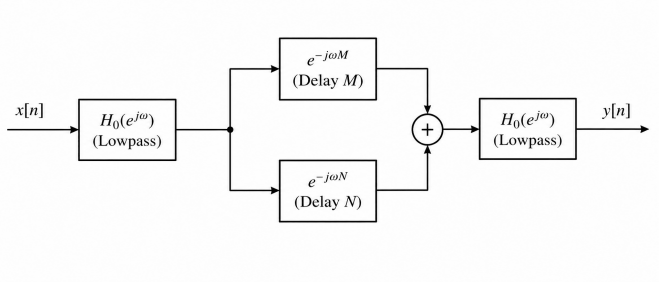


Fig. Q1.14.2

If the BER of this system is $Q(b\sqrt{\gamma})$, then the value of b is _____.

Q1.14.3 NAT 2014 2M

The power spectral density of a real stationary random process $X(t)$ is given by

$$S_X(f) = \begin{cases} \frac{1}{W}, & |f| \leq W \\ 0, & |f| > W \end{cases}$$

The value of the expectation

$$E\left[\pi X(t)\, X\left(t - \frac{1}{4W}\right)\right]$$

is _____.

Q1.14.4 MCQ 2014 2M

Let $X(t)$ be a wide sense stationary (WSS) random process with power spectral density $S_X(f)$. If $Y(t)$ is the process defined as $Y(t) = X(2t - 1)$, the power spectral density $S_Y(f)$ is

A. $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j\pi f}$ **B.** $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j\pi f/2}$
C. $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)$ **D.** $S_Y(f) = \frac{1}{2}S_X\left(\frac{f}{2}\right)e^{-j2\pi f}$

Q1.14.5 NAT 2014 2M

A real band-limited random process $X(t)$ has two-sided power spectral density

$$S_X(f) = \begin{cases} 10^{-6}(3000 - |f|) \text{ Watts/Hz}, & \text{for } |f| \leq 3 \text{ kHz} \\ 0, & \text{otherwise} \end{cases}$$

where f is the frequency expressed in Hz.The signal $X(t)$ modulates a carrier $\cos(16000\pi t)$ and the resultant signal is passed through an ideal band-pass filter of unity gain with centre frequency of 8 kHz and band-width of 2 kHz. The output power (in Watts) is _____.

Q1.14.6 NAT 2014 2M

Let X_1, X_2, and X_3 be independent and identically distributed random variables with the uniform distribution on $[0, 1]$. The probability $P\{X_1 + X_2 \leq X_3\}$ is _____.

Q1.14.7 MCQ 2014 2M

Let X be a real-valued random variable with $E[X]$ and $E[X^2]$ denoting the mean values of X and X^2, respectively. The relation which always holds true is

A. $(E[X])^2 > E[X^2]$ **B.** $E[X^2] \geq (E[X])^2$
C. $E[X^2] = (E[X])^2$ **D.** $E[X^2] > (E[X])^2$

Common Data Questions

The next 2 questions share the data below.

Answer the question 1.13.1 and question 1.13.2 using the following data. Bits 1 and 0 are transmitted with equal probability.At the receiver, the pdf of the respective received signals for both bits are as shown below.

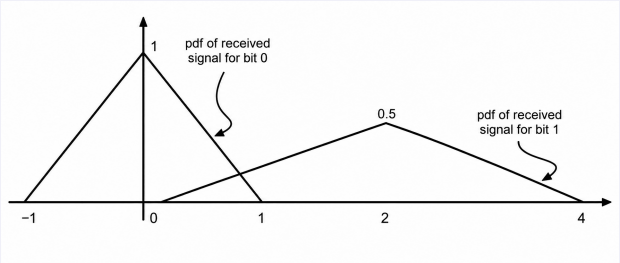


Fig. Q1.14.7

Q1.13.1 MCQ 2013 2M

If the detection threshold is 1, the BER will be

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$
C. $\frac{1}{8}$ **D.** $\frac{1}{16}$

Q1.13.2 MCQ 2013 2M

The optimum threshold to achieve minimum bit error rate (BER) is

A. $\frac{1}{2}$ **B.** $\frac{4}{5}$
C. 1 **D.** $\frac{3}{2}$

Q1.13.3 MCQ 2013 2M

Consider two identically distributed zero-mean random variables U and V. Let the cumulative distribution functions of U and $2V$ be $F(x)$ and $G(x)$ respectively. Then, for all values of x

A. $F(x) - G(x) \leq 0$ **B.** $F(x) - G(x) \geq 0$
C. $(F(x) - G(x))\, x \leq 0$ **D.** $(F(x) - G(x))\, x \geq 0$

Q1.13.4 MCQ 2013 2M

Let U and V be two independent zero mean Gaussian random variables of variances $\frac{1}{4}$ and $\frac{1}{9}$ respectively. The probability $P(3V \geq 2U)$ is

A. $\frac{4}{9}$ **B.** $\frac{1}{2}$
C. $\frac{2}{3}$ **D.** $\frac{5}{9}$

Q1.12.1 MCQ 2012 1M

The power spectral density of a real process $X(t)$ for positive frequencies is shown below. The values of $E[X^2(t)]$ and $E[|X(t)|]$, respectively, are

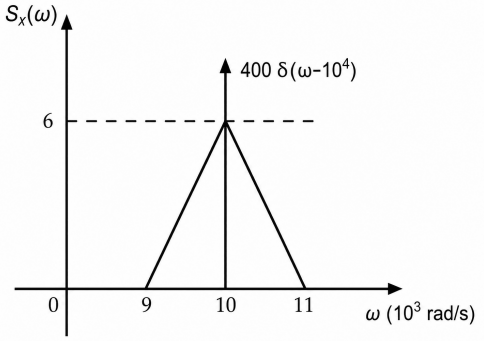


Fig. Q1.12.1

A. $6000/\pi,\ 0$ **B.** $6400/\pi,\ 0$
C. $6400/\pi,\ 20/(\pi\sqrt{2})$ **D.** $6000/\pi,\ 20/(\pi\sqrt{2})$

Q1.12.2 MCQ 2012 1M

Two independent random variables X and Y are uniformly distributed in the interval $[-1, 1]$. The probability that $\max[X, Y]$ is less than $1/2$ is

A. $3/4$ **B.** $9/16$
C. $1/4$ **D.** $2/3$

Q1.11.1 MCQ 2011 2M

$X(t)$ is a stationary random process with autocorrelation function

$$R_X(\tau) = \exp(-\pi\tau^2).$$

This process is passed through the system shown below. The power spectral density of the output process $Y(t)$ is

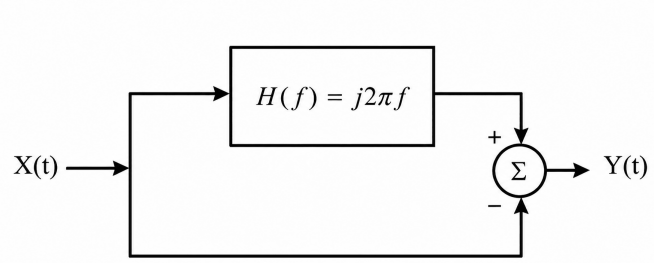


Fig. Q1.11.1

A. $(4\pi^2 f^2 + 1)\exp(-\pi f^2)$ **B.** $(4\pi^2 f^2 - 1)\exp(-\pi f^2)$
C. $(4\pi^2 f^2 + 1)\exp(-\pi f)$ **D.** $(4\pi^2 f^2 - 1)\exp(-\pi f)$

Q1.10.1 MCQ 2010 2M

$X(t)$ is a stationary process with the power spectral density $S_X(f) > 0$ for all f. The process is passed through a system shown below.

Let $S_Y(f)$ be the power spectral density of $Y(t)$. Which one of the following statements is correct?

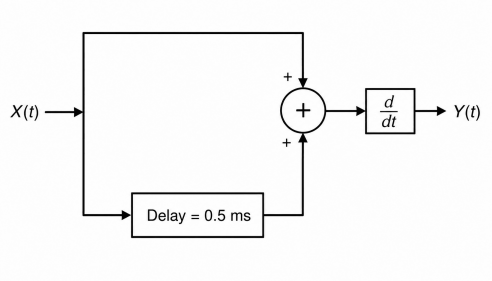


Fig. Q1.10.1

A. $S_Y(f) > 0$ for all f

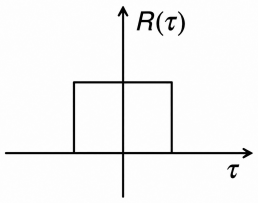
B. $S_Y(f) = 0$ for $|f| > 1$ kHz

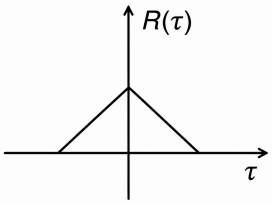
C. $S_Y(f) = 0$ for $f = nf_0$, $f_0 = 2$ kHz, n any integer

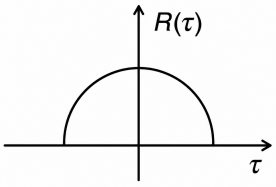
D. $S_Y(f) = 0$ for $f = (2n + 1)f_0$, $f_0 = 1$ kHz, n any integer

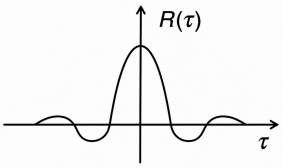
Q1.09.1 MCQ 2009 1M

If the power spectral density of a stationary random process is a sinc-squared function of frequency, the shape of its autocorrelation is _____.

A.


B.


C.


D.


Q1.09.2 MCQ 2009 1M

A white noise process $X(t)$ with two-sided power spectral density

$$S_X(f) = 1 \times 10^{-10} \text{ W/Hz}$$

is input to a filter whose magnitude-squared response $|H(f)|^2$ is triangular, having

$$|H(0)|^2 = 1,$$

and extending from -10 kHz $\le f \le 10$ kHz. The power of the output process $Y(t)$ is

A. 5×10^{-7} W
B. 1×10^{-6} W
C. 2×10^{-6} W
D. 1×10^{-5} W

Q1.08.1 MCQ 2008 2M

Noise with double-sided power spectral density of K over all frequencies is passed through a RC low pass filter with 3 dB cut-off frequency of f_c. The noise power at the filter output is

A. K
B. Kf_c
C. $K\pi f_c$
D. ∞

Q1.08.2 MCQ 2008 1M

The probability density function (PDF) of a random variable X is as shown below.

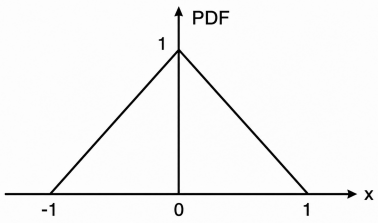
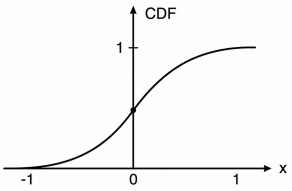
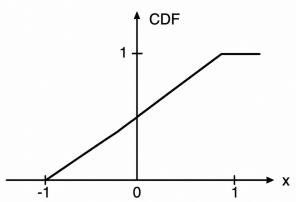
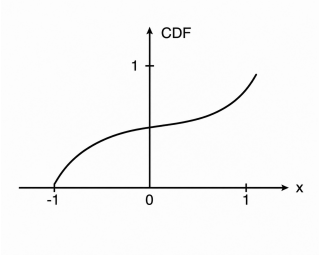


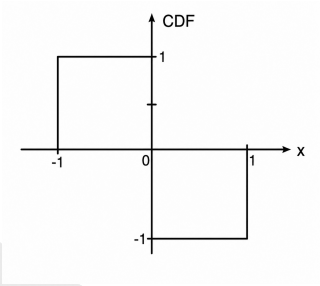
Fig. Q1.08.2

The corresponding cumulative distribution function (CDF) has the form

A.


B.


C.


D.


Q1.07.1 MCQ 2007 1M

If $S(f)$ is the power spectral density of a real, wide-sense stationary random process, then which of the following is ALWAYS true?

A. $S(0) \geq S(f)$
B. $S(f) \geq 0$
C. $S(-f) = -S(f)$
D. $\int_{-\infty}^{\infty} S(f)\, df = 0$

Q1.07.2 MCQ 2007 1M

If $R(\tau)$ is the autocorrelation function of a real, wide-sense stationary random process, then which of the following is NOT true?

A. $R(\tau) = R(-\tau)$
B. $|R(\tau)| \leq R(0)$
C. $R(\tau) = -R(-\tau)$
D. The mean square value of the process is $R(0)$

Q1.06.1 MCQ 2006 2M

A zero-mean white Gaussian noise is passed through an ideal lowpass filter of bandwidth 10 kHz. The output is then uniformly sampled with sampling period $t_s = 0.03$ msec. The samples so obtained would be

A. correlated
B. statistically independent
C. uncorrelated
D. orthogonal

Common Data Questions

The next 2 questions share the data below.

Answer the question 1.06.2 and question 1.06.3 using the following data.
The following two questions refer to the generation of a specified Power Spectral Density (PSD) using an LTI

system/filter.

Q1.06.2 MCQ 2006 2M

It is desired to generate a stochastic process (as voltage process) with power spectral density

$$S(\omega) = \frac{16}{16 + \omega^2}$$

by driving a Linear-Time-Invariant system by zero mean white noise (as voltage process) with power spectral density being constant equal to 1. The system which can perform the desired task could be

A. first order lowpass R-L filter

B. first order highpass R-C filter

C. tuned L-C filter

D. series R-L-C filter

Q1.06.3 MCQ 2006 2M

The parameters of the system obtained in above question would be

A. first order R-L lowpass filter would have
$R = 4\Omega,\ L = 1H$

B. first order R-C highpass filter would have
$R = 4\Omega,\ C = 0.25F$

C. tuned L-C filter would have $L = 4H,\ C = 4F$

D. series R-L-C lowpass filter would have
$R = 1\Omega,\ L = 4H,\ C = 4F$

Q1.06.4 MCQ 2006 2M

A uniformly distributed random variable X with probability density function

$$f_X(x) = \frac{1}{10}\,(u(x+5) - u(x-5))$$

where $u(\cdot)$ is the unit step function, is passed through a transformation given in the figure below. The probability density function of the transformed random variable Y would be

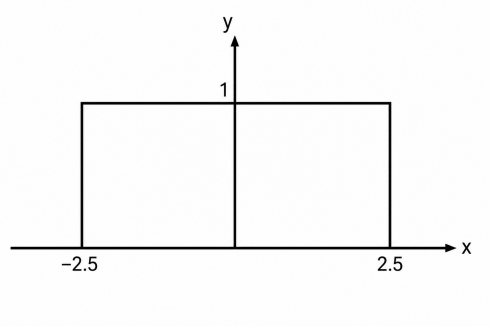


Fig. Q1.06.4

A. $f_Y(y) = \frac{1}{5}\,[u(y+2.5) - u(y-2.25)]$

B. $f_Y(y) = 0.5\,\delta(y) + 0.5\,\delta(y-1)$

C. $f_Y(y) = 0.25\,\delta(y+2.5) + 0.25\,\delta(y-2.5) + 0.5\,\delta(y)$

D. $f_Y(y) = 0.25\,\delta(y+2.5) + 0.25\,\delta(y-2.5) + \frac{1}{10}\,[u(y+2.5) - u(y-2.5)]$

Q1.06.5 MCQ 2006 1M

A probability density function is of the form

$$p(x) = Ke^{-\alpha|x|}, \qquad x \in (-\infty, \infty).$$

The value of K is

A. 0.5 **B.** 1
C. 0.5α **D.** α

Q1.05.1 MCQ 2005 2M

Noise with uniform power spectral density of $N_0 W/\text{Hz}$ is passed through a filter

$$H(\omega) = 2\exp(-j\omega t_d)$$

followed by an ideal low pass filter of bandwidth B Hz. The output noise power in Watts is

A. $2N_0B$ **B.** $4N_0B$
C. $8N_0B$ **D.** $16N_0B$

Q1.05.2 MCQ 2005 2M

An output of a communication channel is a random variable v with the probability density function as shown in figure. The mean square value of v is

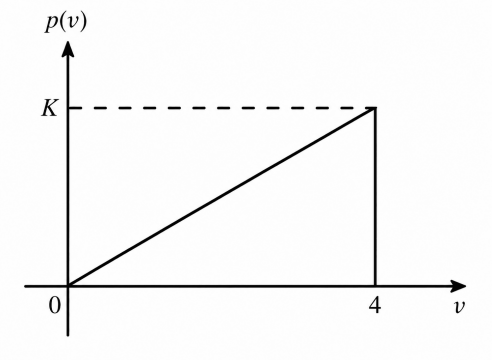


Fig. Q1.05.2

A. 4 **B.** 6
C. 8 **D.** 9

Q1.04.1 MCQ 2004 2M

A random variable X with uniform density in the interval 0 to 1 is quantized as follows:

$$\text{If } 0 \le X \le 0.3, \quad x_q = 0$$
$$\text{If } 0.3 < X \le 1, \quad x_q = 0.7$$

where x_q is the quantized value of X.
The root-mean square value of the quantization noise is

A. 0.573 **B.** 0.198
C. 2.205 **D.** 0.266

Q1.04.2 MCQ 2004 2M

The distribution function $F_X(x)$ of a random variable X is shown in figure below. The probability that $X = 1$ is

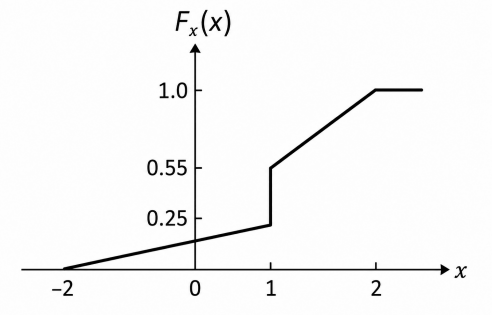


Fig. Q1.04.2

A. zero **B.** 0.25
C. 0.55 **D.** 0.30

Common Data Questions

The next 2 questions share the data below.

Answer the question 1.03.1 and question 1.03.2 using the following data.
$X(t)$ is a random process with a constant mean value of 2 and the auto-correlation function

$$R_X(\tau) = 4\left(e^{-0.2|\tau|} + 1\right).$$

Q1.03.1 MCQ 2003 2M

Let Y and Z be the random variables obtained by sampling $X(t)$ at $t = 2$ and $t = 4$ respectively. Let

$$W = Y - Z.$$

The variance of W is

A. 13.36 **B.** 9.36
C. 2.64 **D.** 8.00

Q1.03.2 MCQ 2003 2M

Let X be the Gaussian random variable obtained by sampling the process at $t = t_i$ and let

$$Q(\alpha) = \int_{\alpha}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-y^2/2}\, dy.$$

The probability that $X \le 1$ is

A. $1 - Q(0.5)$ **B.** $Q(0.5)$
C. $Q\left(\frac{1}{2\sqrt{2}}\right)$ **D.** $1 - Q\left(\frac{1}{2\sqrt{2}}\right)$

Q1.03.3 MCQ 2003 2M

Let X and Y be two statistically independent random variables uniformly distributed in the ranges $(-1, 1)$ and $(-2, 1)$ respectively. Let $Z = X + Y$. Then the probability that $(Z \le -2)$ is

A. zero **B.** $\frac{1}{6}$
C. $\frac{1}{3}$ **D.** $\frac{1}{12}$

Q1.03.4 MCQ 2003 2M

The noise at the input to an ideal frequency detector is white. The detector is operating above threshold. The power spectral density of the noise at the output is

A. raised-cosine **B.** flat
C. parabolic **D.** Gaussian

Q1.02.1 MCQ 2002 2M

If the variance σ_d^2 of

$$d(n) = x(n) - x(n-1)$$

is one-tenth the variance σ_x^2 of a stationary zero mean discrete time signal $x(n)$, then the normalized autocorrelation function

$$\frac{R_{xx}(k)}{\sigma_x^2}$$

at $k = 1$ is

A. 0.95 **B.** 0.90
C. 0.10 **D.** 0.05

Q1.01.1 MCQ 2001 1M

The PDF of a Gaussian random variable X is given by

$$P_X(x) = \frac{1}{3\sqrt{2\pi}} e^{-\frac{(x-4)^2}{18}}.$$

The probability of the event $\{X = 4\}$ is

A. $\frac{1}{2}$ **B.** $\frac{1}{3\sqrt{2\pi}}$
C. 0 **D.** $\frac{1}{4}$

Q1.98.1 MCQ 1998 1M

The ACF of a rectangular pulse of duration T is

A. a rectangular pulse of duration T

B. a rectangular pulse of duration $2T$

C. a triangular pulse of duration T

D. a triangular pulse of duration $2T$

Q1.98.2 MCQ 1998 1M

The amplitude spectrum of a Gaussian pulse is

A. uniform **B.** a sine function
C. Gaussian **D.** an impulse function

Q1.98.3 NAT 1998 2M

White noise of two-sided spectral density

$$2 \times 10^{-6}\ \text{V}^2/\text{Hz}$$

is applied to a simple R-C low-pass filter whose 3 dB cut-off frequency is 4 kHz. Find the mean squared value of the noise output _____ $10^{-2}\ \text{V}^2$.

Q1.97.1 MCQ 1997 1M

A probability density function is given by

$$p(x) = Ke^{-x^2/2}, \qquad -\infty < x < \infty.$$

The value of K should be

A. $\frac{1}{\sqrt{2\pi}}$ **B.** $\sqrt{\frac{2}{\pi}}$
C. $\frac{1}{2\sqrt{\pi}}$ **D.** $\frac{1}{\pi\sqrt{2}}$

Q1.97.2 MCQ 1997 1M

The power spectral density of a deterministic signal is given by

$$\left[\frac{\sin(f)}{f}\right]^2$$

where f is frequency. The autocorrelation function of this signal in the time domain is

A. a rectangular pulse **B.** a delta function
C. a sine pulse **D.** a triangular pulse

Q1.95.1 MCQ 1995 1M

For a narrow band noise with Gaussian quadrature and inphase components, the probability density function of its envelope will be

A. uniform **B.** Gaussian
C. exponential **D.** Rayleigh

Q1.92.1 MCQ 1992 2M

For a random variable X following the probability density function, $p(x)$, shown in figure, the mean and the variance are, respectively.

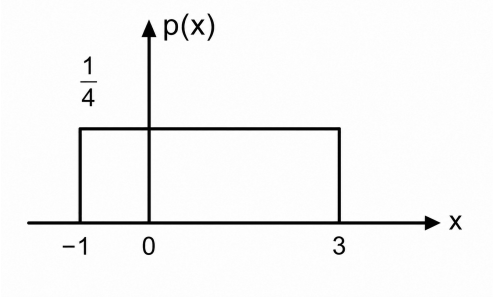


Fig. Q1.92.1

A. 1/2 and 2/3 **B.** 1 and 4/3
C. 1 and 2/3 **D.** 2 and 4/3

Q1.89.1 MCQ 1989 2M

Zero mean Gaussian noise of variance N is applied to a half wave rectifier. The mean squared value of the rectifier output will be:

A. Zero
B. $N/2$
C. $N/\sqrt{2}$
D. N

Q1.87.1 MCQ 1987 2M

The variance of a random variable X is σ_x^2. Then the variance of $-kx$ (where k is a positive constant) is

A. σ_x^2
B. $-k\sigma_x^2$
C. $k\sigma_x^2$
D. $k^2\sigma_x^2$

Answer Key

Q1.26.1 A	Q1.25.1 A	Q1.24.1 A
Q1.24.2 0.5	Q1.23.1 A,B	Q1.23.2 B
Q1.23.3 0.25	Q1.22.1 1.17	Q1.21.1 2.0
Q1.20.1 6	Q1.19.1 17.40 to 17.60	Q1.19.2 11
Q1.19.3 0.36 to 0.38	Q1.18.1 B	Q1.18.2 0.25
Q1.17.1 A	Q1.17.2 2.0	Q1.16.1 -1.01 to -0.99

Q1.16.2 99 to 101	Q1.16.3 B	Q1.15.1 A
Q1.15.2 B	Q1.15.3 D	Q1.15.4 0.24 to 0.26
Q1.14.1 D	Q1.14.2 1.4 to 1.42	Q1.14.3 3.9 to 4.1
Q1.14.4 C	Q1.14.5 2.4 to 2.6	Q1.14.6 0.15 to 0.18
Q1.14.7 B	Q1.13.1 D	Q1.13.2 B
Q1.13.3 D	Q1.13.4 B	Q1.12.1 B

Q1.12.2 B	Q1.11.1 A	Q1.10.1 D
Q1.09.1 B	Q1.09.2 B	Q1.08.1 C
Q1.08.2 A	Q1.07.1 B	Q1.07.2 C
Q1.06.1 A	Q1.06.2 A	Q1.06.3 A
Q1.06.4 B	Q1.06.5 C	Q1.05.1 B
Q1.05.2 C	Q1.04.1 B	Q1.04.2 D
Q1.03.1 C	Q1.03.2 B	Q1.03.3 D
Q1.03.4 C	Q1.02.1 A	Q1.01.1 C
Q1.98.1 D	Q1.98.2 C	Q1.98.3 2.51
Q1.97.1 A	Q1.97.2 D	Q1.95.1 D
Q1.92.1 B	Q1.89.1 B	Q1.87.1 D

UNIT

II

Digital Communications

Topics

1. PCM and DPCM
2. Digital Modulation - ASK, PSK, FSK, QAM
3. Bandwidth and Inter-Symbol Interference
4. MAP and ML Detection
5. Matched Filter Receiver
6. SNR and BER Analysis
7. Error Correction - Hamming Codes and CRC

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 137	**MCQ:** 99	**MSQ:** 0	**NAT:** 38
1M: 41	**2M:** 96	**Descriptive:** 0	**Avg Weightage** ~5M	**Range:** 1M(2023)–8M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q2.26.1 NAT 2026 1M

A wireless digital transmission scheme is using 16-QAM over an additive white Gaussian noise channel and a maximum-likelihood receiver. Consider the information bit rate from source to be 4×10^6 bits per second.
The minimum transmission bandwidth (in MHz) of the modulated signal necessary for optimum recovery of information at the receiver is _____.
(rounded off to two decimal places)

Q2.26.2 MCQ 2026 1M

Consider a discrete memoryless source with an alphabet of four source symbols.
$s(t)$ is a multi-level $(-1, 0, +1, +2)$ signal representing a long sequence of random symbols from the above source which is generating 10^4 symbols per second.
Which of the following options is the correct value of equivalent Nyquist bandwidth of $s(t)$?

A. 10 kHz **B.** 64 kHz
C. 5 kHz **D.** 20 kHz

Q2.26.3 MCQ 2026 2M

A QPSK modulated signal from an additive white Gaussian noise (AWGN) channel is received with an $E_b/N_0 = 8.4$ dB at the input of a coherent QPSK demodulator. A maximum-likelihood reception method is used in the demodulator.
Assume the complementary error function

$$\text{erfc}(u) \cong \left[\frac{1}{u\sqrt{\pi}}\right] \exp(-u^2).$$

Which is the nearest bit error rate (BER) at the output of the demodulator?

A. 10^{-3} **B.** 10^{-4}
C. 10^{-5} **D.** 10^{-6}

Q2.26.4 NAT 2026 2M

The average bit error rate at the input of a $(7, 4, 1)$ Hamming decoder is 0.10.
The probability that the decoder will fail to decode a received word correctly is _____.
(rounded off to two decimal places)

Q2.25.1 NAT 2025 1M

The generator matrix of a $(6, 3)$ binary linear block code is given by

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{bmatrix}$$

The minimum Hamming distance $d_{\min}$ between codewords equals _____.
(answer in integer)

Q2.25.2 MCQ 2025 2M

A source transmits symbol S that takes values uniformly at random from the set $\{-2, 0, 2\}$. The receiver obtains $Y = S + N$, where N is a zero-mean Gaussian random variable independent of S. The receiver uses the maximum likelihood decoder to estimate the transmitted symbol S.
Suppose the probability of symbol estimation error P_e is expressed as follows:

$$P_e = \alpha\, P(N > 1),$$

where $P(N > 1)$ denotes the probability that N exceeds 1.What is the value of α?

A. $1/3$ **B.** 1
C. $2/3$ **D.** $4/3$

Q2.24.1 MCQ 2024 1M

A digital communication system transmits through a noiseless bandlimited channel $[-W\ W]$. The received signal $z(t)$ at the output of the receiving filter is given by $z(t) = \sum_n b[n]x(t - nT)$ where $b[n]$ are the symbols and $x(t)$ is the overall system response to a single symbol. The received signal is sampled at $t = mT$. The Fourier transform of $x(t)$ is $X(f)$. The Nyquist condition that $X(f)$ must satisfy for zero intersymbol interference at the receiver is _____.

A. $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = T$

B. $\sum_{m=-\infty}^{\infty} X\left(f + \frac{m}{T}\right) = \frac{1}{T}$

C. $\sum_{m=-\infty}^{\infty} X(f + mT) = T$

D. $\sum_{m=-\infty}^{\infty} X(f + mT) = \frac{1}{T}$

Q2.24.2 MCQ 2024 2M

A source transmits a symbol s, taken from $\{-4, 0, 4\}$ with equal probability, over an additive white Gaussian noise channel. The received noisy symbol r is given by $r = s + w$, where the noise w is zero mean with variance 4 and is independent of s.

Using

$$Q(x) = \frac{1}{\sqrt{2\pi}} \int_x^{\infty} e^{-t^2/2}\, dt,$$

the optimum symbol error probability is _____.

A. $\frac{2}{3}Q(2)$ **B.** $\frac{4}{3}Q(1)$

C. $\frac{2}{3}Q(1)$ **D.** $\frac{4}{3}Q(2)$

Q2.24.3 MCQ 2024 2M

The information bit sequence {111010101} is to be transmitted by encoding with Cyclic Redundancy Check 4 (CRC-4) code, for which the generator polynomial is $C(x) = x^4 + x + 1$.
The encoded sequence of bits is _____.

A. {1110101011100}

B. {1110101011101}

C. {1110101011110}

D. {1110101010100}

Q2.24.4 NAT 2024 2M

Let $X(t) = A\cos(2\pi f_0 t + \theta)$ be a random process, where amplitude A and phase θ are independent of each other, and are uniformly distributed in the intervals $[-2, 2]$ and $[0, 2\pi]$, respectively. $X(t)$ is fed to an 8-bit uniform mid-rise type quantizer. Given that the autocorrelation of $X(t)$ is $R_X(\tau) = \frac{2}{3}\cos(2\pi f_0 \tau)$,the signal to quantization noise ratio (in dB, rounded off to two decimal places) at the output of the quantizer is _____.

Q2.23.1 NAT 2023 1M

Let the signal-to-noise ratio (SNR) of an ADC with a full-scale sinusoidal input be given as 61.96 dB.
The resolution of the ADC is _____ bits (rounded to the nearest integer).

Q2.22.1 NAT 2022 1M

A symbol stream contains alternate QPSK and 16-QAM symbols. If symbols from this stream are transmitted at the rate of 1 mega-symbols per second, the raw (uncoded) data rate is _____ mega-bits per second.
(rounded off to one decimal place)

Q2.22.2 MCQ 2022 1M

The frequency response $H(f)$ of a linear time-invariant system has magnitude as shown in the figure.
Statement I : The system is necessarily a pure delay system for inputs which are bandlimited to $-\alpha \le f \le \alpha$.
Statement II : For any wide-sense stationary input process with power spectral density $S_X(f)$, the output power spectral density $S_Y(f)$ obeys $S_Y(f) = S_X(f)$ for $-\alpha \le f \le \alpha$.

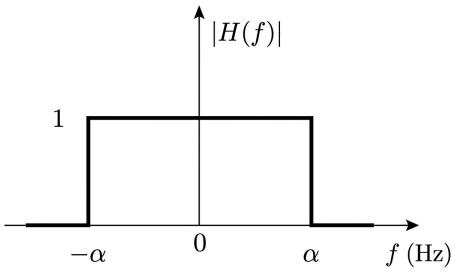


Fig. Q2.22.2

Which one of the following combinations is true?

A. Statement I is correct, Statement II is correct

B. Statement I is correct, Statement II is incorrect

C. Statement I is incorrect, Statement II is correct

D. Statement I is incorrect, Statement II is incorrect

Q2.22.3 NAT 2022 2M

Consider communication over a memoryless binary symmetric channel using a $(7, 4)$ Hamming code. Each transmitted bit is received correctly with probability $(1-\epsilon)$, and flipped with probability ϵ. For each codeword transmission, the receiver performs minimum Hamming distance decoding, and correctly decodes the message bits if and only if the channel introduces at most one bit error.
For $\epsilon = 0.1$, the probability that a transmitted codeword is decoded correctly is _____.
(rounded off to two decimal places)

Q2.22.4 NAT 2022 2M

Consider a channel over which either symbol x_A or symbol x_B is transmitted. Let the output of the channel Y be the input to a maximum likelihood (ML) detector at the receiver. The conditional probability density functions for Y given x_A and x_B are:

$$f_{Y|x_A}(y) = e^{-(y+1)}u(y+1),$$

$$f_{Y|x_B}(y) = e^{(y-1)}\big(1 - u(y-1)\big),$$

where $u(\cdot)$ is the standard unit step function. The probability of symbol error for this system is _____. (rounded off to two decimal places)

Q2.21.1 NAT 2021 1M

A speech signal, band limited to 4 kHz, is sampled at 1.25 times the Nyquist rate. The speech samples, assumed to be statistically independent and uniformly distributed in the range −5 V to +5 V, are subsequently quantized in an 8-bit uniform quantizer and then transmitted over a voice-grade AWGN telephone channel. If the ratio of transmitted signal power to channel noise power is 26 dB, the minimum channel bandwidth required to ensure reliable transmission of the signal with arbitrarily small probability of transmission error _____ kHz.(rounded off to two decimal places)

Q2.21.2 NAT 2021 1M

A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a delta modulator (DM) operating at a sampling rate of 32 kHz. The minimum step size required to avoid slope overload noise in the DM _____ V.(rounded off to two decimal places)

Q2.21.3 MCQ 2021 2M

A digital transmission system uses a $(7, 4)$ systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-codeword pairs in this code $(m_i; c_i)$, where c_i is the codeword corresponding to the i^{th} message m_i, are known to be $(1100; 0101100)$, $(1110; 0011110)$, $(0110; 1000110)$, then which of the following is a valid codeword in this code?

A. 1101001 **B.** 1011010
C. 0001011 **D.** 0110100

Q2.21.4 NAT 2021 2M

A message signal having peak-to-peak value of 2 V, root mean square value of 0.1 V and bandwidth of 5 kHz is sampled and fed to a pulse code modulation (PCM) system that uses a uniform quantizer. The PCM output is transmitted over a channel that can support a maximum transmission rate of 50 kbps. Assuming that the quantization error is uniformly distributed, the maximum signal to quantization noise ratio that can be obtained by the PCM system _____.(rounded off to two decimal places and calculate answer in linear scale)

Q2.21.5 NAT 2021 2M

A polar non-return to zero (NRZ) waveform, using +2 V and −2 V for representing binary '1' and '0' respectively, is transmitted in the presence of additive zero-mean white Gaussian noise with variance 0.4 V^2. If the *a priori* probability of transmission of a binary '1' is 0.4, the optimum threshold voltage for a maximum *a posteriori* (MAP) receiver _____ V.(rounded off to two decimal places)

Q2.20.1 MCQ 2020 1M

A digital communication system transmits a block of N bits. The probability of error in decoding a bit is α. The error event of each bit is independent of the error events of the other bits. The received block is declared erroneous if at least one of its bits is decoded wrongly. The probability that the received block is erroneous is

A. $N(1-\alpha)$ **B.** α^N
C. $1-\alpha^N$ **D.** $1-(1-\alpha)^N$

Q2.20.2 NAT 2020 2M

In a digital communication system, a symbol S randomly chosen from the set $\{s_1, s_2, s_3, s_4\}$ is transmitted. It is given that $s_1 = -3$, $s_2 = -1$,$s_3 = +1$, and $s_4 = +2$. The received symbol is $Y = S + W$. W is a zero-mean unit-variance Gaussian random variable and is independent of S. P_i is the conditional probability of symbol error for the maximum likelihood (ML) decoding when the transmitted symbol $S = s_i$. The index i for which the conditional symbol error probability P_i is the highest is _____.

Q2.19.1 MCQ 2019 1M

A linear Hamming code is used to map 4-bit messages to 7-bit codewords. The encoder mapping is linear. If the message 0001 is mapped to the codeword 0000111, and the message 0011 is mapped to the codeword 1100110, then the message 0010 is mapped to

A. 0010011 **B.** 1100001
C. 1111000 **D.** 1111111

Q2.19.2 MCQ 2019 2M

A single bit, equally likely to be 0 and 1, is to be sent across an additive white Gaussian noise (AWGN) channel with power spectral density $N_0/2$. Binary signaling, with $0 \mapsto p(t)$ and $1 \mapsto q(t)$, is used for the transmission, along with an optimal receiver that minimizes the bit-error probability.

Let $\varphi_1(t), \varphi_2(t)$ form an orthonormal signal set.

If we choose $p(t) = \varphi_1(t)$ and $q(t) = -\varphi_1(t)$, we would obtain a certain bit-error probability P_b.

If we keep $p(t) = \varphi_1(t)$, but take $q(t) = \sqrt{E}\,\varphi_2(t)$, for what value of E would we obtain the same bit-error probability P_b?

A. 0 **B.** 1
C. 2 **D.** 3

Q2.19.3 NAT 2019 2M

A voice signal $m(t)$ is in the frequency range 5 kHz to 15 kHz. The signal is amplitude-modulated to generate an AM signal $f(t) = A(1 + m(t)) \cos 2\pi f_c t$, where $f_c = 600$ kHz. The AM signal $f(t)$ is to be digitized and archived. This is done by first sampling $f(t)$ at 1.2 times the Nyquist frequency, and then quantizing each sample using a 256-level quantizer. Finally, each quantized sample is binary coded using K bits, where K is the minimum number of bits required for the encoding. The rate, in Megabits per second (rounded off to 2 decimal places), of the resulting stream of coded bits is _____ Mbps.

Q2.19.4 NAT 2019 2M

A random variable X takes values -1 and $+1$ with probabilities 0.2 and 0.8, respectively. It is transmitted across a channel which adds noise N, so that the random variable at the channel output is $Y = X + N$. The noise N is independent of X, and is uniformly distributed over the interval $[-2, 2]$. The receiver makes a decision

$$\hat{X} = \begin{cases} -1, & \text{if } Y \le \theta \\ +1, & \text{if } Y > \theta \end{cases}$$

where the threshold $\theta \in [-1, 1]$ is chosen so as to minimize the probability of error $\Pr[\hat{X} \neq X]$. The minimum probability of error, rounded off to 1 decimal place, is _____.

Q2.18.1 NAT 2018 1M

Consider a binary channel code in which each codeword has a fixed length of 5 bits. The Hamming distance between any pair of distinct codewords in this code is at least 2. The maximum number of codewords such a code can contain is _____.

Q2.18.2 NAT 2018 1M

A binary source generates symbols $X \in \{-1, 1\}$ which are transmitted over a noisy channel. The probability of transmitting $X = 1$ is 0.5. Input to the threshold detector is $R = X + N$. The probability density function $f_N(n)$ of the noise N is shown below.

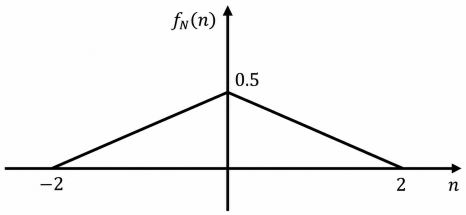


Fig. Q2.18.2

If the detection threshold is zero, then the probability of error (correct to two decimal places) is _____.

Q2.18.3 NAT 2018 2M

A random variable X takes values -0.5 and 0.5 with probabilities $\frac{1}{4}$ and $\frac{3}{4}$, respectively. The noisy observation of X is $Y = X + Z$, where Z has uniform probability density over the interval $(-1, 1)$. X and Z are independent. If the MAP rule based detector outputs $\hat{X}$ as

$$\hat{X} = \begin{cases} -0.5, & Y < \alpha \\ 0.5, & Y \ge \alpha, \end{cases}$$

then the value of α (accurate to two decimal places) is _____.

Q2.17.1 NAT 2017 1M

A sinusoidal message signal is converted to a PCM signal using a uniform quantizer. The required signal-to-quantization noise ratio (SQNR) at the output of the quantizer is 40 dB. The minimum number of bits per sample needed to achieve the desired SQNR is _____.

Q2.17.2 MCQ 2017 1M

Which one of the following statements about differential pulse code modulation (DPCM) is true?

A. The sum of message signal sample with its prediction is quantized

B. The message signal sample is directly quantized, and its prediction is not used

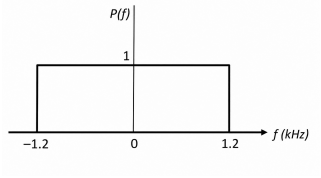
C. The difference of message signal sample and a random signal is quantized

D. The difference of message signal sample with its prediction is quantized

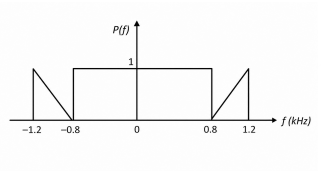
Q2.17.3 MCQ 2017 1M

In a digital communication system, the overall pulse shape $p(t)$ at the receiver before the sampler has the Fourier transform $P(f)$. If the symbols are transmitted at the rate of 2000 symbols per second, for which of the following cases is the inter symbol interference zero?

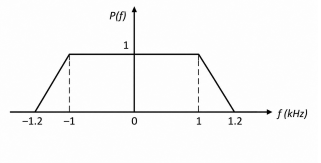
A.



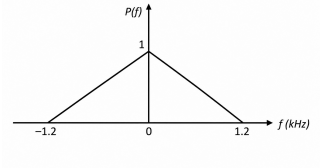
B.



C.



D.



Q2.17.4 MCQ 2017 2M

In binary frequency shift keying (FSK), the given signal waveforms are

$$u_0(t) = 5\cos(20000\pi t);\ 0 \le t \le T,$$

and

$$u_1(t) = 5\cos(22000\pi t);\ 0 \le t \le T,$$

where T is the bit-duration interval and t is in seconds. Both $u_0(t)$ and $u_1(t)$ are zero outside the interval $0 \le t \le T$. With a matched filter (correlator) based receiver, the smallest positive value of T (in milliseconds) required to have $u_0(t)$ and $u_1(t)$ uncorrelated is

A. 0.25 ms **B.** 0.5 ms
C. 0.75 ms **D.** 1.0 ms

Q2.16.1 NAT 2016 1M

Consider binary data transmission at a rate of 56 kbps using baseband binary pulse amplitude modulation (PAM) that is designed to have a raised-cosine spectrum. The transmission bandwidth (in kHz) required for a roll-off factor of 0.25 is _____.

Q2.16.2 NAT 2016 1M

A discrete memoryless source has an alphabet a_1,a_2,a_3, a_4 with corresponding probabilities $\left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8}\right\}$.The minimum required average codeword length in bits to represent this source for error-free reconstruction is _____.

Q2.16.3 NAT 2016 1M

A speech signal is sampled at 8 kHz and encoded into PCM format using 8 bits/sample. The PCM data is transmitted through a baseband channel via 4-level PAM. The minimum bandwidth (in kHz) required for transmission is _____.

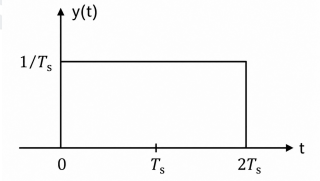
Q2.16.4 MCQ 2016 1M

A binary baseband digital communication system employs the signal

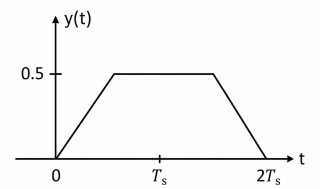
$$p(t) = \begin{cases} \frac{1}{\sqrt{T_s}}, & 0 \le t \le T_s \\ 0, & \text{otherwise} \end{cases}$$

for transmission of bits. The graphical representation of the matched filter output $y(t)$ for this signal will be

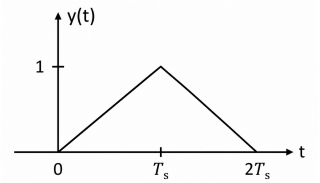
A.



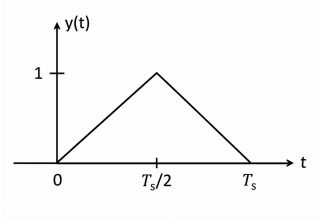
B.



C.



D.



Q2.16.5 NAT 2016 2M

A voice-grade AWGN (additive white Gaussian noise) telephone channel has a bandwidth of 4.0 kHz and two-sided noise power spectral density $\frac{\eta}{2} = 2.5 \times 10^{-5}$ Watt per Hz.If information at the rate of 52 kbps is to be transmitted over this channel with arbitrarily small bit error rate, then the minimum bit-energy E_b (in mJ/bit) necessary is _____.

Q2.16.6 NAT 2016 2M

The bit error probability of a memoryless binary symmetric channel is 10^{-5}. If 10^5 bits are sent over this channel, then the probability that not more than one bit will be in error is _____.

Q2.16.7 NAT 2016 2M

An ideal band-pass channel 500 Hz - 2000 Hz is deployed for communication. A modem is designed to transmit bits at the rate of 4800 bits/s using 16-QAM. The roll-off factor of a pulse with a raised cosine spectrum that utilizes the entire frequency band is _____.

Q2.16.8 NAT 2016 2M

A digital communication system uses a repetition code for channel encoding/decoding. During transmission, each bit is repeated three times (0 is transmitted as 000, and 1 is transmitted as 111). It is assumed that the source puts out symbols independently and with equal probability. The decoder operates as follows: In a block of three received bits, if the number of zeros exceeds the number of ones, the decoder decides in favor of a 0, and if the number of ones exceeds the number of zeros, the decoder decides in favor of a 1. Assuming a binary symmetric channel with crossover probability $p = 0.1$, the average probability of error is _____.

Q2.16.9 MCQ 2016 2M

An analog pulse $s(t)$ is transmitted over an additive white Gaussian noise (AWGN) channel. The received signal is $r(t) = s(t) + n(t)$, where $n(t)$ is additive white Gaussian noise with power spectral density $N_0/2$. The received signal is passed through a filter with impulse response $h(t)$. Let E_s and E_h denote the energies of the pulse $s(t)$ and the filter $h(t)$, respectively. When the signal-to-noise ratio (SNR) is maximized at the output of the filter ($\text{SNR}_{\max}$), which of the following holds?

A. $E_s = E_h;\ \text{SNR}_{\max} = \dfrac{2E_s}{N_0}$

B. $E_s = E_h;\ \text{SNR}_{\max} = \dfrac{E_s}{2N_0}$

C. $E_s > E_h;\ \text{SNR}_{\max} > \dfrac{2E_s}{N_0}$

D. $E_s < E_h;\ \text{SNR}_{\max} = \dfrac{2E_h}{N_0}$

Q2.15.1 NAT 2015 2M

The input X to the Binary Symmetric Channel (BSC) shown in the figure is '1' with probability 0.8. The crossover probability is $\frac{1}{7}$. If the received bit $Y = 0$, the conditional probability that '1' was transmitted is _____.

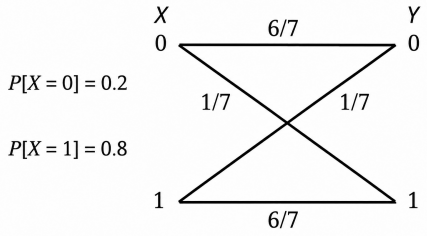


Fig. Q2.15.1

Q2.15.2 MCQ 2015 2M

A source emits bit 0 with probability $\frac{1}{3}$ and bit 1 with probability $\frac{2}{3}$. The emitted bits are communicated to the receiver. The receiver decides for either 0 or 1 based on the received value R. It is given that the conditional density functions of R are as

$$f_{R|0}(r) = \begin{cases} \frac{1}{4}, & -3 \le x \le 1, \\ 0, & \text{otherwise}, \end{cases}$$

$$f_{R|1}(r) = \begin{cases} \frac{1}{6}, & -1 \le x \le 5, \\ 0, & \text{otherwise}. \end{cases}$$

The minimum decision error probability is.

A. 0 **B.** $\frac{1}{12}$

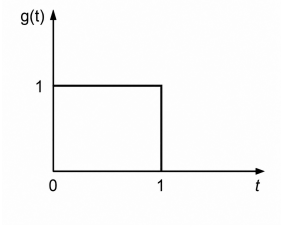
C. $\frac{1}{9}$ **D.** $\frac{1}{6}$

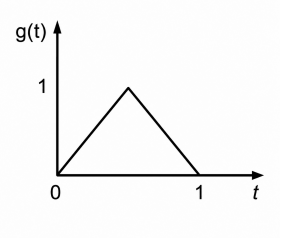
Q2.15.3 NAT 2015 2M

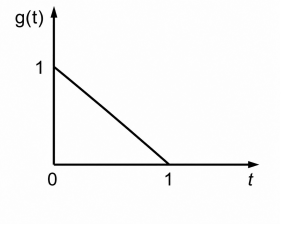
The transmitted signal in a GSM system is of 200 kHz bandwidth and 8 users share a common bandwidth using TDMA. If at a given time 12 users are talking in a cell, the total bandwidth of the signal received by the base station of the cell will be at least (in kHz) _____.

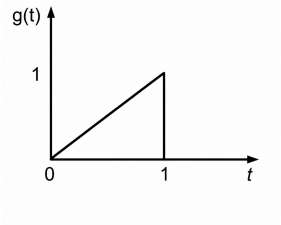
Q2.15.4 MCQ 2015 2M

Consider a binary digital communication system which uses pulses $g(t)$ and $-g(t)$ for transmitting bits over an AWGN channel. If the receiver uses a matched filter, which one of the following pulses will give the minimum probability of bit error?

A.


B.


C.


D.


Q2.15.5 NAT 2015 1M

A sinusoidal signal of amplitude A is quantized by a uniform quantizer. Assume that the signal utilizes all the representation levels of the quantizer. If the signal to quantization noise ratio is 31.8 dB, the number of levels in the quantizer is _____.

Q2.15.6 MCQ 2015 1M

A sinusoidal signal of 2 kHz frequency is applied to a delta modulator. The sampling rate and step-size Δ of the delta modulator are 20, 000 samples per second and 0.1 V, respectively. To prevent slope overload, the maximum amplitude of the sinusoidal signal (in Volts) is

A. $\frac{1}{2\pi}$ **B.** $\frac{1}{\pi}$

C. $\frac{2}{\pi}$ **D.** π

Q2.15.7 MCQ 2015 1M

The modulation scheme commonly used for transmission from GSM mobile terminals is

A. 4-QAM
B. 16-PSK
C. Walsh-Hadamard orthogonal codes
D. Gaussian Minimum Shift Keying (GMSK)

Q2.14.1 NAT 2014 2M

The input to a 1-bit quantizer is a random variable X with pdf

$$f_X(x) = \begin{cases} 2e^{-2x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

For outputs to be of equal probability, the quantizer threshold should be _____.

Q2.14.2 MCQ 2014 2M

Coherent orthogonal binary FSK modulation is used to transmit two equiprobable symbol waveforms $s_1(t) = \alpha \cos 2\pi f_1 t$ and $s_2(t) = \alpha \cos 2\pi f_2 t$, where $\alpha = 4$ mV. Assume an AWGN channel with two-sided noise power spectral density $\frac{N_0}{2} = 0.5 \times 10^{-12}$ W/Hz. Using an optimal receiver and the relation

$$Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du,$$

the bit error probability for a data rate of 500 kbps is

A. $Q(2)$ **B.** $Q(2\sqrt{2})$

C. $Q(4)$ **D.** $Q(4\sqrt{2})$

Q2.14.3 NAT 2014 2M

In a PCM system, the signal

$$m(t) = \{\sin(100\pi t) + \cos(100\pi t)\} \text{ V}$$

is sampled at the Nyquist rate. The samples are processed by a uniform quantizer with step size 0.75 V. The minimum data rate of the PCM system in bits per second is _____.

Q2.14.4 MCQ 2014 2M

Consider a communication scheme where the binary valued signal X satisfies

$$P\{X = +1\} = 0.75 \quad \text{and} \quad P\{X = -1\} = 0.25.$$

The received signal is

$$Y = X + Z,$$

where Z is a Gaussian random variable with zero mean and variance σ^2. The received signal Y is fed to the threshold detector. The output of the threshold detector $\hat{X}$ is

$$\hat{X} = \begin{cases} +1, & Y > \tau, \\ -1, & Y \le \tau. \end{cases}$$

To achieve a minimum probability of error $P\{\hat{X} \neq X\}$, the threshold τ should be

A. strictly positive
B. zero
C. strictly negative
D. strictly positive, zero, or strictly negative depending on the nonzero value of σ^2

Q2.14.5 NAT 2014 2M

An M-level PSK modulation scheme is used to transmit independent binary digits over a band-pass channel with bandwidth 100 kHz. The bit rate is 200 kbps and the system characteristic is a raised-cosine spectrum with 100% excess bandwidth. The minimum value of M is _____.

Q2.14.6 MCQ 2014 2M

Consider a discrete-time channel $Y = X + Z$, where the additive noise Z is signal-dependent. In particular, given the transmitted symbol $X \in \{-a, +a\}$ at any instant, the noise sample Z is chosen independently from a Gaussian distribution with mean βX and unit variance. Assume a threshold detector with zero threshold at the receiver.

When $\beta = 0$, the BER was found to be $Q(a) = 1 \times 10^{-8}$.

$$Q(v) = \frac{1}{\sqrt{2\pi}} \int_v^{\infty} e^{-u^2/2}\, du,$$

and for $v > 1$, use

$$Q(v) \approx e^{-v^2/2}.$$

When $\beta = -0.3$, the BER is closest to

A. 10^{-7} **B.** 10^{-6}
C. 10^{-4} **D.** 10^{-2}

Q2.14.7 NAT 2014 1M

In a code-division multiple access (CDMA) system with $N = 8$ chips, the maximum number of users who can be assigned mutually orthogonal signature sequences is _____.

Q2.14.8 NAT 2014 1M

An analog voltage in the range 0 to 8 V is divided in 16 equal intervals for conversion to a 4-bit digital output. The maximum quantization error (in V) is _____.

Q2.14.9 NAT 2014 1M

A modulated signal is $y(t) = m(t)\cos(40000\pi t)$, where the baseband signal $m(t)$ has frequency components less than 5 kHz only. The minimum required sampling rate (in kHz) at which $y(t)$ should be sampled to recover $m(t)$ is_____.

Q2.13.1 MCQ 2013 1M

Let $g(t) = e^{-\pi t^2}$, and $h(t)$ is a filter matched to $g(t)$. If $g(t)$ is applied as input to $h(t)$, then the Fourier transform of the output is

A. $e^{-\pi f^2}$ **B.** $e^{-\pi f^2/2}$
C. $e^{-\pi |f|}$ **D.** $e^{-2\pi f^2}$

Q2.13.2 MCQ 2013 1M

The bit rate of a digital communication system is R kbits/s. The modulation used is 32-QAM. The minimum bandwidth required for ISI-free transmission is

A. $R/10$ Hz **B.** $R/10$ kHz
C. $R/5$ Hz **D.** $R/5$ kHz

Q2.12.1 MCQ 2012 1M

In a baseband communications link, frequencies upto 3500 Hz are used for signaling. Using a raised cosine pulse with 75% excess bandwidth and for no inter-symbol interference, the maximum possible signaling rate in symbols per second is

A. 1750 **B.** 2625
C. 4000 **D.** 5250

Q2.12.2 MCQ 2012 2M

A BPSK scheme operating over an AWGN channel with noise power spectral density of $N_0/2$, uses equiprobable signals

$$s_1(t) = \sqrt{\frac{2E}{T}}\sin(\omega_c t)$$

and

$$s_2(t) = -\sqrt{\frac{2E}{T}}\sin(\omega_c t)$$

over the symbol interval $(0, T)$. If the local oscillator in a coherent receiver is ahead in phase by 45° with respect to the received signal, the probability of error in the resulting system is

A. $Q\left(\sqrt{\frac{2E}{N_0}}\right)$ **B.** $Q\left(\sqrt{\frac{E}{N_0}}\right)$

C. $Q\left(\sqrt{\frac{E}{2N_0}}\right)$ **D.** $Q\left(\sqrt{\frac{E}{4N_0}}\right)$

Q2.12.3 MCQ 2012 2M

A binary symmetric channel (BSC) has a transition probability of $\frac{1}{8}$. If the binary transmit symbol X is such that

$$P(X = 0) = \frac{9}{10},$$

then the probability of error for an optimum receiver is

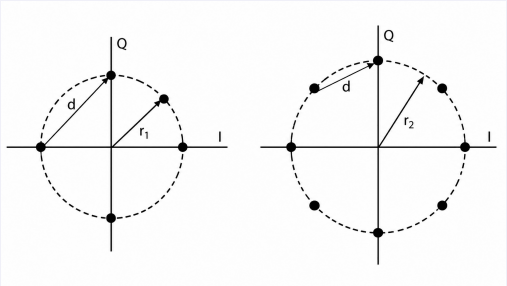
A. $\frac{7}{80}$ **B.** $\frac{63}{80}$

C. $\frac{9}{10}$ **D.** $\frac{1}{10}$

Common Data Questions

The next 2 questions share the data below.

Answer the question 2.11.1 and question 2.11.2 using the following data. A four-phase and an eight-phase signal constellation are shown in the figure below.



Q2.11.1 MCQ 2011 2M

For the constraint that the minimum distance between pairs of signal points be d for both constellations, the radii r_1 and r_2 of the circles are

A. $r_1 = 0.707d,\ r_2 = 2.782d$

B. $r_1 = 0.707d,\ r_2 = 1.932d$

C. $r_1 = 0.707d,\ r_2 = 1.545d$

D. $r_1 = 0.707d,\ r_2 = 1.307d$

Q2.11.2 MCQ 2011 2M

Assuming high SNR and that all signals are equally probable, the additional average transmitted signal energy required by the 8-PSK signal to achieve the same error probability as the 4-PSK signal is

A. 11.90 dB **B.** 8.73 dB

C. 6.79 dB **D.** 5.33 dB

Q2.11.3 MCQ 2011 1M

An analog signal is band-limited to 4 kHz, sampled at the Nyquist rate and the samples are quantized into 4 levels. The quantized levels are assumed to be independent and equally probable. If we transmit two quantized samples per second, the information rate is

A. 1 bit/sec **B.** 2 bits/sec

C. 3 bits/sec **D.** 4 bits/sec

Common Data Questions

The next 2 questions share the data below.

Answer the question 2.10.1 and question 2.10.2 using the following data. Consider a baseband binary PAM receiver shown below. The additive channel noise $n(t)$ is white with power spectral density

$$S_N(f) = \frac{N_0}{2} = 10^{-20}\ \text{W/Hz}.$$

The low-pass filter is ideal with unity gain and cutoff frequency 1 MHz. Let Y_k represent the random variable $y(t_k)$.

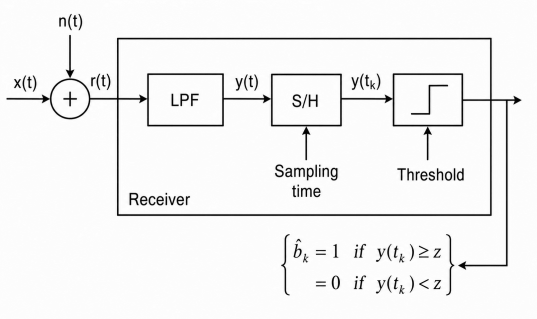
$$Y_k = \begin{cases} N_k, & \text{if transmitted bit } b_k = 0, \\ a + N_k, & \text{if transmitted bit } b_k = 1, \end{cases}$$

where N_k represents the noise sample value. The noise sample has a probability density function

$$p_{N_k}(n) = 0.5\,\alpha e^{-\alpha|n|}.$$

(This has mean zero and variance $2/\alpha^2$.) Assume transmitted bits to be equiprobable and threshold τ is set to

$$\tau = \frac{a}{2} = 10^{-6} \text{ V}.$$



Q2.10.1 MCQ 2010 2M

The value of the parameter α (in V^{-1}) is

A. 10^{10} **B.** 10^{7}
C. 1.414×10^{-10} **D.** 2×10^{20}

Q2.10.2 MCQ 2010 2M

The probability of bit error is

A. $0.5 \times e^{-0.5}$ **B.** $0.5 \times e^{-5}$
C. $0.5 \times e^{-7}$ **D.** $0.5 \times e^{-10}$

Q2.10.3 MCQ 2010 1M

Consider the pulse shape $s(t)$ as shown. The impulse response $h(t)$ of the filter matched to this pulse is

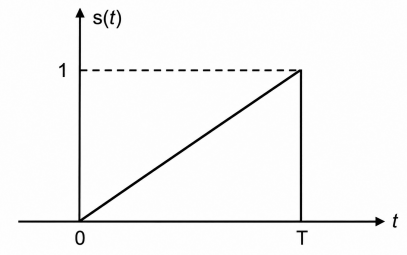
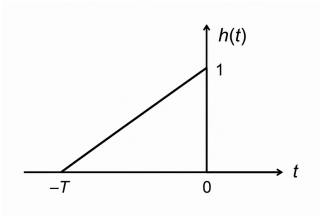
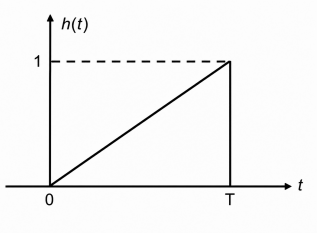
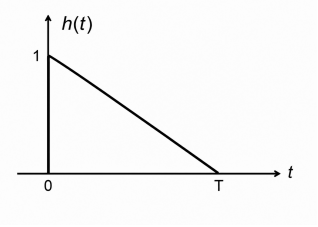
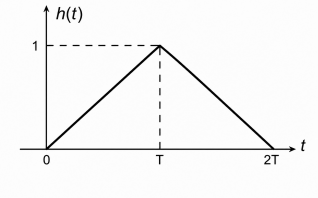


Fig. Q2.10.3

A.


B.


C.


D.


Common Data Questions

The next 2 questions share the data below.

Answer the question 2.09.1 and question 2.09.2 using the following data. The amplitude of a random signal is uniformly distributed between −5 V and 5 V.

Q2.09.1 MCQ 2009 2M

If the signal to quantization noise ratio required in uniformly quantizing the signal is 43.5 dB, the step size of the quantization is approximately

A. 0.0333 V **B.** 0.05 V
C. 0.0667 V **D.** 0.10 V

Q2.09.2 MCQ 2009 2M

If the positive values of the signal are uniformly quantized with a step size of 0.05 V, and the negative values are uniformly quantized with a step size of 0.1 V, the resulting signal to quantization noise ratio is approximately

A. 46 dB **B.** 43.8 dB
C. 42 dB **D.** 40 dB

Q2.08.1 MCQ 2008 2M

Four messages band limited to W, W, $2W$ and $3W$ respectively are to be multiplexed using Time Division Multiplexing (TDM). The minimum bandwidth required for transmission of this TDM signal is

A. W **B.** $3W$
C. $6W$ **D.** $7W$

Common Data Questions

The next 3 questions share the data below.

Answer the questions 2.08.2 , 2.08.3 , 2.08.4 using the following data.
A speech signal, band limited to 4 kHz and peak voltage varying between +5 V and −5 V, is sampled at the Nyquist rate. Each sample is quantized and represented by 8 bits.

Q2.08.2 MCQ 2008 2M

If the bits 0 and 1 are transmitted using bipolar pulses, the minimum bandwidth required for distortion free transmission is

A. 64 kHz **B.** 32 kHz
C. 8 kHz **D.** 4 kHz

Q2.08.3 MCQ 2008 2M

Assuming the signal to be uniformly distributed between its peak-to-peak value, the signal to noise ratio at the quantizer output is

A. 16 dB **B.** 32 dB
C. 48 dB **D.** 64 dB

Q2.08.4 MCQ 2008 2M

The number of quantization levels required to reduce the quantization noise by a factor of 4 would be

A. 1024 **B.** 512
C. 256 **D.** 64

Q2.07.1 MCQ 2007 2M

In delta modulation, the slope overload distortion can be reduced by

A. decreasing the step size
B. decreasing the granular noise
C. decreasing the sampling rate
D. increasing the step size

Q2.07.2 MCQ 2007 2M

The raised cosine pulse $p(t)$ is used for zero ISI in digital communications. The expression for $p(t)$ with unity roll-off factor is given by

$$p(t) = \frac{\sin 4\pi Wt}{4\pi Wt\left(1 - 16W^2t^2\right)}.$$

The value of $p(t)$ at

$$t = \frac{1}{4W}$$

is

A. -0.5 **B.** 0
C. 0.5 **D.** ∞

Q2.07.3 MCQ 2007 2M

In a GSM system, 8 channels can co-exist in 200 kHz bandwidth using TDMA. A GSM based cellular operator is allocated 5 MHz bandwidth. Assuming a frequency reuse factor of $\frac{1}{5}$, i.e. a five-cell repeat pattern, the maximum number of simultaneous channels that can exist in one cell is

A. 200 **B.** 40
C. 25 **D.** 5

Q2.07.4 MCQ 2007 2M

In a Direct Sequence CDMA system the chip rate is 1.2288×10^6 chips per second. If the processing gain is desired to be AT LEAST 100, the data rate

A. must be less than or equal to 12.288×10^3 bits per sec
B. must be greater than 12.288×10^3 bits per sec
C. must be exactly equal to 12.288×10^3 bits per sec
D. can take any value less than 122.88×10^3 bits per sec

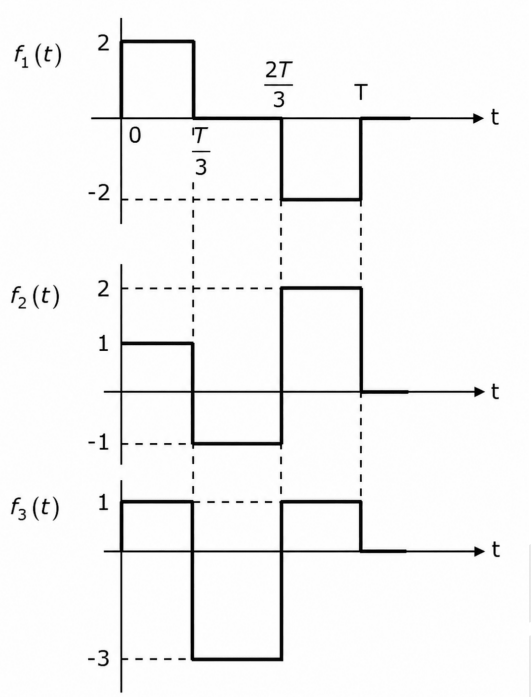
Q2.07.5 MCQ 2007 2M

During transmission over a certain binary communication channel, bit errors occur independently with probability p. The probability of AT MOST one bit in error in a block of n bits is given by

A. p^n **B.** $1 - p^n$
C. $np(1-p)^{n-1} + (1-p)^n$ **D.** $1 - (1-p)^n$

Q2.07.6 MCQ 2007 2M

Three functions $f_1(t)$, $f_2(t)$ and $f_3(t)$, which are zero outside the interval $[0, T]$, are shown in the figure. Which of the following statements is correct?

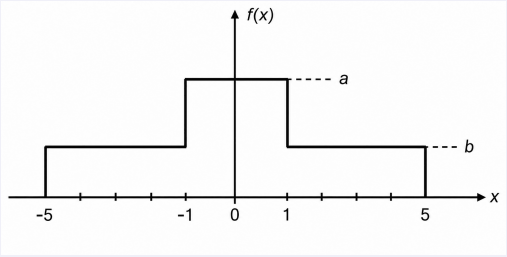


A. $f_1(t)$ and $f_2(t)$ are orthogonal
B. $f_1(t)$ and $f_3(t)$ are orthogonal
C. $f_3(t)$ and $f_3(t)$ are orthogonal
D. $f_1(t)$ and $f_2(t)$ are orthonormal

Common Data Questions

The next 2 questions share the data below.

Answer the question 2.07.7 and question 2.07.8 using the following data.
An input to a 6-level quantizer has the probability density function $f(x)$ as shown in the figure. Decision boundaries of the quantizer are chosen so as to maximize the entropy of the quantizer output. It is given that 3 consecutive decision boundaries are -1, 0 and 1.



Q2.07.7 MCQ 2007 2M

The values of a and b are:

A. $a = \frac{1}{6}$ and $b = \frac{1}{12}$
B. $a = \frac{1}{5}$ and $b = \frac{3}{40}$
C. $a = \frac{1}{4}$ and $b = \frac{1}{16}$
D. $a = \frac{1}{3}$ and $b = \frac{1}{24}$

Q2.07.8 MCQ 2007 2M

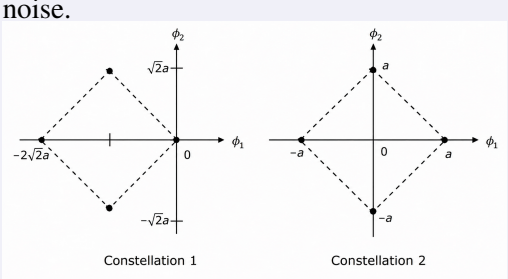
Assuming that the reconstruction levels of the quantizer are the mid-points of the decision boundaries, the ratio of signal power to quantization noise power is

A. $\frac{152}{9}$
B. $\frac{64}{3}$
C. $\frac{76}{3}$
D. 28

Common Data Questions

The next 2 questions share the data below.

Answer the question 2.07.9 and question 2.07.10 using the following data.
Two 4-ary signal constellations are shown. It is given that ϕ_1 and ϕ_2 constitute an orthonormal basis for the two constellations. Assume that the four symbols in both the constellations are equiprobable. Let $\frac{N_0}{2}$ denote the power spectral density of white Gaussian noise.



Q2.07.9 MCQ 2007 2M

The ratio of the average energy of Constellation 1 to the average energy of Constellation 2 is:

A. $4a^2$
B. 4
C. 2
D. 8

Q2.07.10 MCQ 2007 2M

If these constellations are used for digital communications over an AWGN channel, then which of the following statements is true?
A. Probability of symbol error for Constellation 1 is lower
B. Probability of symbol error for Constellation 1 is higher
C. Probability of symbol error is equal for both the constellations
D. The value of N_0 will determine which of the two constellations has a lower probability of symbol error

Q2.06.1 MCQ 2006 2M

The minimum step-size required for a Delta-Modulator operating at 32 K samples/sec to track the signal (here $u(t)$ is the unit-step function)

$$x(t) = 125t\big(u(t) - u(t-1)\big)$$

$$+\big(250 - 125t\big)\big(u(t-1) - u(t-2)\big)$$

so that slope-overload is avoided, would be

A. 2^{-10} **B.** 2^{-8}
C. 2^{-6} **D.** 2^{-4}

Q2.06.2 MCQ 2006 2M

A source generates three symbols with probabilities 0.25, 0.25, 0.50 at a rate of 3000 symbols per second. Assuming independent generation of symbols, the most efficient source encoder would have average bit rate as

A. 6000 bits/sec **B.** 4500 bits/sec
C. 3000 bits/sec **D.** 1500 bits/sec

Common Data Questions

The next 2 questions share the data below.

Answer the question 2.06.3 and question 2.06.4 using the following data.
Let

$$g(t) = p(t) * p(t),$$

where $*$ denotes convolution and

$$p(t) = u(t) - u(t-1),$$

with $u(t)$ being the unit step function.

Q2.06.3 MCQ 2006 2M

The impulse response of filter matched to the signal

$$s(t) = g(t) - \delta(t-2) * g(t)$$

is given as:

A. $s(1-t)$ **B.** $-s(1-t)$
C. $-s(t)$ **D.** $s(t)$

Q2.06.4 MCQ 2006 2M

In the following figure the minimum value of the constant "C", which is to be added to $y_1(t)$ such that $y_1(t)$ and $y_2(t)$ are different, is

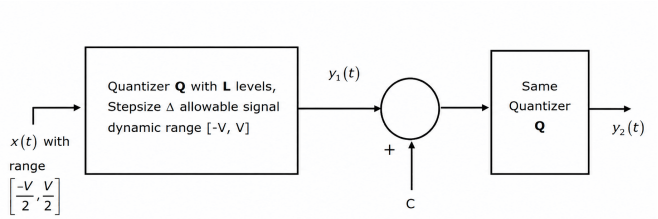


Fig. Q2.06.4

A. Δ **B.** $\frac{\Delta}{2}$
C. $\frac{\Delta^2}{12}$ **D.** $\frac{\Delta}{L}$

Q2.06.5 MCQ 2006 1M

The minimum sampling frequency (in samples/sec) required to reconstruct the following signal from its samples without distortion,

$$x(t) = 5\left(\frac{\sin 2\pi 1000t}{\pi t}\right)^3 + 7\left(\frac{\sin 2\pi 1000t}{\pi t}\right)^2$$

would be

A. 2×10^3 **B.** 4×10^3
C. 6×10^3 **D.** 8×10^3

Q2.05.1 MCQ 2005 2M

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels. The quantization noise power for the quantization region between $-a$ and $+a$ in the figure is

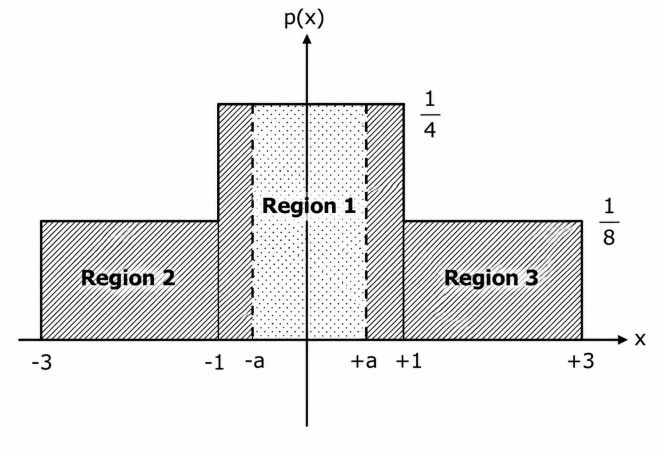


Fig. Q2.05.1

A. $\frac{4}{81}$

B. $\frac{1}{9}$

C. $\frac{5}{81}$

D. $\frac{2}{81}$

Q2.05.2 MCQ 2005 2M

A symmetric three-level midtread quantizer is to be designed assuming equiprobable occurrence of all quantization levels. If the probability density function is divided into three regions as shown in the figure, the value of a in the figure is _____.

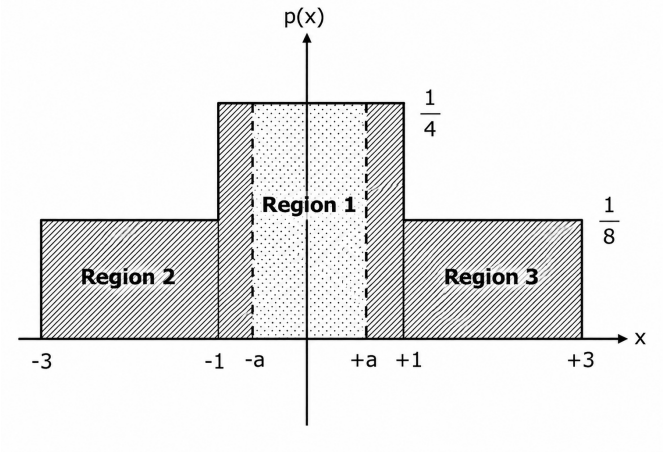


Fig. Q2.05.2

A. $\frac{1}{3}$

B. $\frac{2}{3}$

C. $\frac{1}{2}$

D. $\frac{1}{4}$

Q2.05.3 MCQ 2005 2M

A Signal as shown in the figure is applied to a matched filter.Which of the following does represent the output of this matched filter?

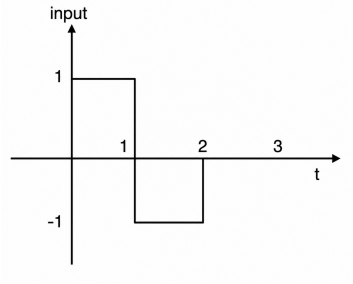
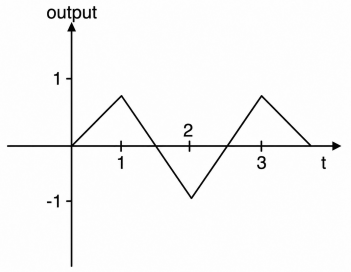
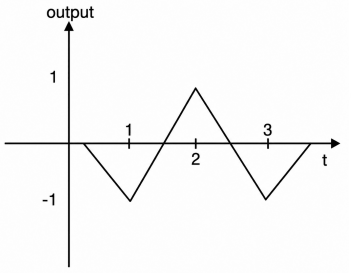


Fig. Q2.05.3

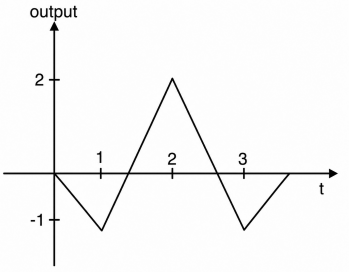
A.



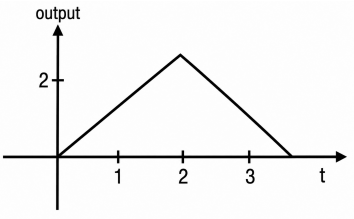
B.



C.



D.



Q2.04.1 MCQ 2004 2M

A source produces binary data at the rate of 10 kbps. The binary symbols are represented as shown in the figure given below.

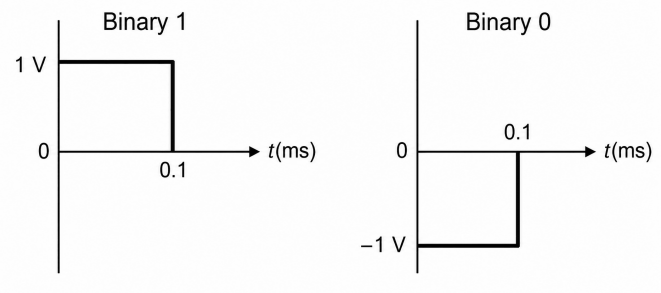


Fig. Q2.04.1

The source output is transmitted using two modulation schemes, namely Binary PSK (BPSK) and Quadrature PSK (QPSK). Let B_1 and B_2 be the bandwidth requirements of BPSK and QPSK respectively. Assume that the bandwidth of the above rectangular pulses is 10 kHz. B_1 and B_2 are

A. $B_1 = 20$ kHz, $B_2 = 20$ kHz
B. $B_1 = 10$ kHz, $B_2 = 20$ kHz
C. $B_1 = 20$ kHz, $B_2 = 10$ kHz
D. $B_1 = 10$ kHz, $B_2 = 10$ kHz

Q2.04.2 MCQ 2004 2M

Three analog signals, having bandwidths 1200 Hz, 600 Hz and 600 Hz, are sampled at their respective Nyquist rates, encoded with 12 bit words, and time division multiplexed. The bit rate for the multiplexed signal is

A. 115.2 kbps **B.** 28.8 kbps
C. 57.6 kbps **D.** 38.4 kbps

Q2.04.3 MCQ 2004 2M

Consider a binary digital communication system with equally likely 0's and 1's. When binary 0 is transmitted, the detector input can lie between the levels −0.25 V and +0.25 V with equal probability; when binary 1 is transmitted, the voltage at the detector can have any value between 0 and 1 V with equal probability. If the detector has a threshold of 0.2 V (i.e., if the received signal is greater than 0.2 V, the bit is taken as 1), the average bit error probability is

A. 0.15 **B.** 0.2
C. 0.05 **D.** 0.5

Q2.04.4 MCQ 2004 2M

A 1 mW video signal having a bandwidth of 100 MHz is transmitted to a receiver through a cable that has 40 dB loss. If the effective one-side noise spectral density at the receiver is 10^{-20} Watt/Hz, then the signal-to-noise ratio at the receiver is

A. 50 dB **B.** 30 dB
C. 40 dB **D.** 60 dB

Q2.04.5 MCQ 2004 2M

Consider the signal $x(t)$ shown in Fig. Let $h(t)$ denote the impulse response of the filter matched to $x(t)$, with $h(t)$ being non-zero only in the interval 0 to 4 sec. The slope of $h(t)$ in the interval $3 < t < 4$ sec is

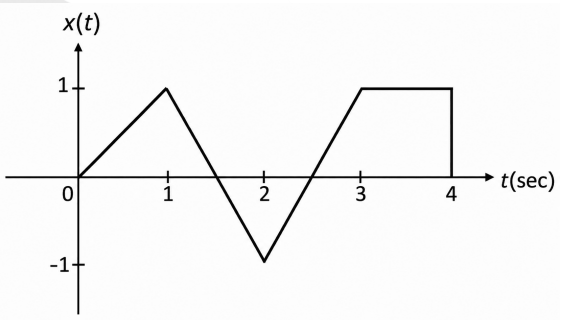


Fig. Q2.04.5

A. $\frac{1}{2}$ sec^{-1} **B.** -1 sec^{-1}
C. $-\frac{1}{2}$ sec^{-1} **D.** 1 sec^{-1}

Q2.04.6 MCQ 2004 2M

In the output of a DM speech encoder, the consecutive pulses are of opposite polarity during time interval $t_1 \leq t \leq t_2$. This indicates that during this interval

A. the input to the modulator is essentially constant
B. the modulator is going through slope overload
C. the accumulator is in saturation
D. the speech signal is being sampled at the Nyquist rate

Q2.04.7 MCQ 2004 2M

In a PCM system, if the code word length is increased from 6 to 8 bits, the signal to quantization noise ratio improves by the factor

A. $\frac{8}{6}$ **B.** 12
C. 16 **D.** 8

Q2.03.1 MCQ 2003 2M

A signal is sampled at 8 kHz and is quantized using an 8-bit uniform quantizer. Assuming SNR_q for a sinusoidal signal, the correct statement for PCM signal with a bit rate of R is

A. $R = 32$ kbps, $\text{SNR}_q = 25.8$ dB
B. $R = 64$ kbps, $\text{SNR}_q = 49.8$ dB
C. $R = 64$ kbps, $\text{SNR}_q = 55.8$ dB
D. $R = 32$ kbps, $\text{SNR}_q = 49.8$ dB

Q2.03.2 MCQ 2003 2M

If S represents the carrier synchronization at the receiver and ρ represents the bandwidth efficiency, then the correct statement for the coherent binary PSK is

A. $\rho = 0.5$, S is required
B. $\rho = 1.0$, S is required
C. $\rho = 0.5$, S is not required
D. $\rho = 1.0$, S is not required

Q2.03.3 MCQ 2003 2M

The input to a linear delta modulator having a step-size $\Delta = 0.628$ is a sine wave with frequency f_m and peak amplitude E_m. If the sampling frequency is $f_s = 40$ kHz,the combination of the sine-wave frequency and the peak amplitude, where slope overload will take place, is

A. $E_m = 0.3$ V, $f_m = 8$ kHz
B. $E_m = 1.5$ V, $f_m = 4$ kHz
C. $E_m = 1.5$ V, $f_m = 2$ kHz
D. $E_m = 0.3$ V, $f_m = 1$ kHz

Q2.03.4 MCQ 2003 2M

If E_b, the energy per bit of a binary digital signal, is 10^{-5} watt-sec and the one-sided power spectral density of the white noise, $N_0 = 10^{-6}$ W/Hz, then the output SNR of the matched filter is

A. 26 dB **B.** 10 dB
C. 20 dB **D.** 13 dB

Q2.03.5 MCQ 2003 2M

A sinusoidal signal with peak-to-peak amplitude of 1.536 V is quantized into 128 levels using a mid-rise uniform quantizer. The quantization-noise power is

A. 0.768 V **B.** 48×10^{-6} V^2
C. 12×10^{-6} V^2 **D.** 3.072 V

Q2.03.6 MCQ 2003 2M

At a given probability of error, binary coherent FSK is inferior to binary coherent PSK by

A. 6 dB **B.** 3 dB
C. 2 dB **D.** 0 dB

Q2.02.1 MCQ 2002 2M

A signal

$$x(t) = 100\cos\left(24\pi \times 10^3\, t\right)$$

is ideally sampled with a sampling period of 50 μsec and then passed through an ideal low-pass filter with cutoff frequency of 15 kHz. Which of the following frequencies is/are present at the filter output?

A. 12 kHz only
B. 8 kHz only
C. 12 kHz and 9 kHz
D. 12 kHz and 8 kHz

Q2.02.2 MCQ 2002 2M

For a bit-rate of 8 Kbps, the best possible values of the transmitted frequencies in a coherent binary FSK system are

A. 16 kHz and 20 kHz
B. 20 kHz and 32 kHz
C. 20 kHz and 40 kHz
D. 32 kHz and 40 kHz

Q2.02.3 MCQ 2002 2M

Consider a sampled signal

$$y(t) = 5 \times 10^{-6}\, x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$$

where

$$x(t) = 10\cos(8\pi \times 10^3\, t)$$

and

$$T_s = 100\,\mu\text{sec}.$$

When $y(t)$ is passed through an ideal low-pass filter with a cutoff frequency of 5 kHz, the output of the filter is

A. $5 \times 10^{-6}\cos(8\pi \times 10^3\, t)$ **B.** $5 \times 10^{-5}\cos(8\pi \times 10^3\, t)$
C. $5 \times 10^{-1}\cos(8\pi \times 10^3\, t)$ **D.** $10\cos(8\pi \times 10^3\, t)$

Q2.01.1 MCQ 2001 2M

The Nyquist sampling interval, for the signal

$$\text{sinc}(700t) + \text{sinc}(500t)$$

is

A. $\frac{1}{350}$ sec **B.** $\frac{\pi}{350}$ sec

C. $\frac{1}{700}$ sec **D.** $\frac{\pi}{175}$ sec

Q2.01.2 MCQ 2001 2M

A video transmission system transmits 625 picture frames per second. Each frame consists of a 400×400 pixel grid with 64 intensity levels per pixel. The data rate of the system is

A. 16 Mbps **B.** 100 Mbps

C. 600 Mbps **D.** 6.4 Gbps

Q2.00.1 MCQ 2000 2M

In a digital communication system employing Frequency Shift Keying (FSK), the 0 and 1 bit are represented by sine waves of 10 kHz and 25 kHz respectively. These waveforms will be orthogonal for a bit interval of

A. 45 μsec **B.** 200 μsec

C. 50 μsec **D.** 250 μsec

Q2.99.1 MCQ 1999 2M

The peak-to-peak input to an 8-bit PCM encoder is 2 volts. The signal power-to-quantization noise power ratio (in dB) for an input of $0.5\cos(\omega_m t)$ is

A. 47.8 **B.** 49.8

C. 95.6 **D.** 99.6

Q2.99.2 MCQ 1999 2M

The input to a matched filter is given by

$$s(t) = \begin{cases} 10\sin(2\pi \times 10^6 t), & 0 < t < 10^{-4} \text{ sec} \\ 0, & \text{otherwise} \end{cases}$$

The peak amplitude of the filter output is

A. 10 volts **B.** 5 volts

C. 10 millivolts **D.** 5 millivolts

Q2.99.3 MCQ 1999 2M

Four independent messages have bandwidths of 100 Hz, 100 Hz, 200 Hz and 400 Hz respectively. Each is sampled at the Nyquist rate, and the samples are time division multiplexed (TDM) and transmitted. The transmitted sample rate (in Hz) is

A. 1600 **B.** 800

C. 400 **D.** 200

Q2.98.1 MCQ 1998 1M

In a PCM system with uniform quantization, increasing the number of bits from 8 to 9 will reduce the quantization noise power by a factor of

A. 9 **B.** 8

C. 4 **D.** 2

Q2.98.2 MCQ 1998 1M

Quadrature multiplexing is

A. the same as FDM

B. the same as TDM

C. a combination of FDM and TDM

D. quite different from FDM and TDM

Q2.98.3 MCQ 1998 1M

Compression in PCM refers to relative compression of

A. higher signal amplitudes

B. lower signal amplitudes

C. lower signal frequencies

D. higher signal frequencies

Q2.98.4 MCQ 1998 1M

For a given data rate, the bandwidth B_P of a BPSK signal and the bandwidth B_0 of the OOK signal are related as

A. $B_P = \frac{B_0}{4}$ **B.** $B_P = \frac{B_0}{2}$

C. $B_P = B_0$ **D.** $B_P = 2B_0$

Q2.97.1 MCQ 1997 1M

The line code that has zero dc component for pulse transmission of random binary data is

A. Non-return to zero (NRZ)

B. Return to zero (RZ)

C. Alternate Mark Inversion (AMI)

D. None of the above

Q2.97.2 MCQ 1997 1M

A deterministic signal has the power spectrum given in the figure. The minimum sampling rate needed to completely represent this signal is

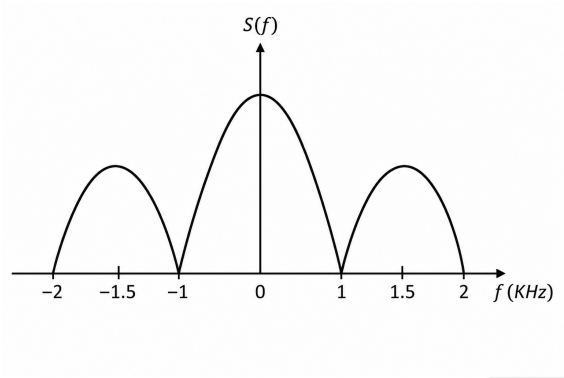


Fig. Q2.97.2

A. 1 kHz **B.** 2 kHz
C. 3 kHz **D.** None of the above

Q2.96.1 MCQ 1996 2M

The number of bits in a binary PCM system is increased from n to $n+1$. As a result, the signal-to-quantization-noise ratio will improve by a factor

A. $\frac{n+1}{n}$ **B.** $2^{\frac{n+1}{n}}$
C. $2^{\frac{2(n+1)}{n}}$ **D.** which is independent of n

Q2.95.1 MCQ 1995 1M

If the number of bits per sample in a PCM system is increased from n to $n + 1$, the improvement in signal-to-quantization noise ratio will be

A. 3 dB **B.** 6 dB
C. $2n$ dB **D.** n dB

Q2.95.2 MCQ 1995 1M

The signal-to-quantization noise ratio in an n-bit PCM system

A. depends upon the sampling frequency employed
B. is independent of the value of n
C. increases with increasing value of n
D. decreases with increasing value of n

Q2.94.1 NAT 1994 1M

The bandwidth required for the transmission of a PCM signal increases by a factor of _____ when the number of quantization levels is increased from 4 to 64.

Q2.92.1 MCQ 1992 2M

The bit stream 01001 is differentially encoded using the Delay and Ex-OR scheme for DPSK transmission. Assuming the reference bit as a 1 and assigning phases of 0 and π for 1s and 0s respectively, in the encoded sequence, the transmitted phase sequence becomes

A. $\pi,\ 0,\ \pi,\ \pi,\ 0$ **B.** $0,\ \pi,\ \pi,\ 0,\ 0$
C. $0,\ \pi,\ \pi,\ \pi,\ 0$ **D.** $\pi,\ \pi,\ 0,\ \pi,\ \pi$

Q2.92.2 MCQ 1992 2M

Coherent demodulation of FSK signal can be effected using

A. correlation receiver
B. band-pass filters and envelope detectors
C. matched filter
D. discriminator detection

Q2.92.3 MCQ 1992 2M

Source encoding in a data communication system is done in order to

A. Enhance the information transmission
B. Band-pass filters and envelope rate detectors
C. Conserve the transmitted power
D. Discriminator detection

Q2.91.1 NAT 1991 2M

A binary source has symbol probabilities 0.8 and 0.2. If extension coding (blocks of 4 symbols) is used, the lower and upper bounds on the average code word length are:

(a) lower _____

(b) higher _____

Q2.91.2 NAT 1991 2M

For the signal constellation shown in the figure the type of modulation is

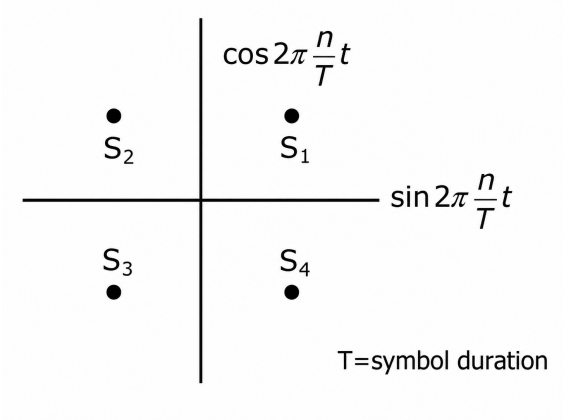


Fig. Q2.91.2

Q2.90.1 MCQ 1990 2M

In a BPSK signal detector, the local oscillator has a fixed phase error of 20°. This phase error deteriorates the output signal-to-noise ratio (SNR) by a factor of

A. $\cos 20°$ **B.** $\cos^2 20°$
C. $\cos 70°$ **D.** $\cos^2 70°$

Q2.89.1 MCQ 1989 2M

In binary data transmission, DPSK is preferred to PSK because

(a) a coherent carrier is not required to be generated at the receiver.
(b) for a given energy per bit, the probability of error is less.
(c) the 180° phase shifts of the carrier are unimportant.
(d) more protection is provided against impulse noise.

Q2.88.1 MCQ 1988 2M

The transfer function of a zero-order hold is

A. $\dfrac{1-\exp(-Ts)}{s}$ **B.** $\dfrac{1}{s}$
C. 1 **D.** $\dfrac{1}{1-\exp(-Ts)}$

Q2.88.2 MCQ 1988 2M

The message bit sequence input to a DPSK modulator is

$$1,\ 1,\ 0,\ 0,\ 1,\ 1.$$

The carrier phase during the reception of the first two message bits is $\pi,\ \pi$. The carrier phase for the remaining four message bits is

A. $\pi,\ \pi,\ 0,\ \pi$ **B.** $0,\ 0,\ \pi,\ \pi$
C. $0,\ \pi,\ \pi,\ \pi$ **D.** $\pi,\ \pi,\ 0,\ 0$

Q2.88.3 MCQ 1988 2M

A signal having uniformly distributed amplitude in the interval $(-V, +V)$ is to be encoded using PCM with uniform quantization. The signal-to-quantization noise ratio is determined by the

A. dynamic range of the signal
B. sampling rate
C. number of quantizing levels
D. power spectrum of signal

Q2.88.4 MCQ 1988 2M

In a digital communication system, transmissions of successive bits through a noisy channel are assumed to be independent events with error probability p. The probability of at most one error in the transmission of an 8-bit sequence is

A. $\dfrac{7(1-p)}{8}+\dfrac{p}{8}$ **B.** $(1-p)^8+8p(1-p)^7$
C. $(1-p)^8+(1-p)^7$ **D.** $(1-p)^8+p(1-p)^7$

Q2.87.1 MCQ 1987 2M

Companding in PCM systems leads to improved signal-to-quantization noise ratio. This improvement is for

A. lower frequency components only
B. higher frequency components only
C. lower amplitudes only
D. higher amplitudes only

Answer Key

Q2.26.1 0.95 to 1.05	Q2.26.2 C	Q2.26.3 B
Q2.26.4 0.10 to 0.20	Q2.25.1 3	Q2.25.2 D
Q2.24.1 A(MTA)	Q2.24.2 B	Q2.24.3 A
Q2.24.4 45.00 to 45.30	Q2.23.1 10	Q2.22.1 2.99 to 3.01
Q2.22.2 C	Q2.22.3 0.84 to 0.86	Q2.22.4 0.23
Q2.21.1 9.24 to 9.28	Q2.21.2 2.80 to 3.20	Q2.21.3 C

Q2.21.4 30.00 to 34.00	Q2.21.5 0.03 to 0.05	Q2.20.1 D
Q2.20.2 3	Q2.19.1 B	Q2.19.2 D
Q2.19.3 11.80 to 11.82	Q2.19.4 0.1	Q2.18.1 16
Q2.18.2 0.12 to 0.14	Q2.18.3 -0.5 to -0.5	Q2.17.1 7
Q2.17.2 D	Q2.17.3 B	Q2.17.4 B
Q2.16.1 34.5 to 35.5	Q2.16.2 1.74 to 1.76	Q2.16.3 15.9 to 16.1
Q2.16.4 C	Q2.16.5 30.0 to 33.0	Q2.16.6 0.70 to 0.75
Q2.16.7 0.24 to 0.26	Q2.16.8 0.025 to 0.030	Q2.16.9 A
Q2.15.1 0.39 to 0.41	Q2.15.2 D	Q2.15.3 400
Q2.15.4 A	Q2.15.5 32	Q2.15.6 A
Q2.15.7 D	Q2.14.1 0.34 to 0.36	Q2.14.2 C
Q2.14.3 199 to 201	Q2.14.4 C	Q2.14.5 15.9 to 16.1
Q2.14.6 C	Q2.14.7 8	Q2.14.8 0.24 to 0.26
Q2.14.9 9.5 to 10.5	Q2.13.1 D	Q2.13.2 D
Q2.12.1 C	Q2.12.2 B	Q2.12.3 D
Q2.11.1 D	Q2.11.2 D	Q2.11.3 D
Q2.10.1 B	Q2.10.2 D	Q2.10.3 C
Q2.09.1 C	Q2.09.2 C	Q2.08.1 D
Q2.08.2 B	Q2.08.3 C	Q2.08.4 B
Q2.07.1 D	Q2.07.2 C	Q2.07.3 B
Q2.07.4 A	Q2.07.5 C	Q2.07.6 C
Q2.07.7 A	Q2.07.8 D	Q2.07.9 B
Q2.07.10 A	Q2.06.1 B	Q2.06.2 B
Q2.06.3 C	Q2.06.4 B	Q2.06.5 C

Q2.05.1 A	Q2.05.2 B	Q2.05.3 C
Q2.04.1 C	Q2.04.2 C	Q2.04.3 A
Q2.04.4 A	Q2.04.5 B	Q2.04.6 A
Q2.04.7 C	Q2.03.1 B	Q2.03.2 B
Q2.03.3 B	Q2.03.4 D	Q2.03.5 C
Q2.03.6 B	Q2.02.1 D	Q2.02.2 A
Q2.02.3 C	Q2.01.1 C	Q2.01.2 C
Q2.00.1 B	Q2.99.1 43.9 (MTA)	Q2.99.2 D
Q2.99.3 A	Q2.98.1 C	Q2.98.2 D
Q2.98.3 A	Q2.98.4 C	Q2.97.1 C
Q2.97.2 B	Q2.96.1 D	Q2.95.1 B
Q2.95.2 C	Q2.94.1 3	Q2.92.1 C
Q2.92.2 A	Q2.92.3 A	Q2.91.1 2.8 , 3.8
Q2.91.2 QPSK	Q2.90.1 B	Q2.89.1 A
Q2.88.1 A	Q2.88.2 C	Q2.88.3 C
Q2.88.4 B	Q2.87.1 C	

UNIT

III

Information Theory

Topics

1. Entropy and Mutual Information
2. Channel Capacity Theorem

Unit Snapshot*

Stream: EC	**Questions:** 24	**MCQ:** 10	**MSQ:** 2	**NAT:** 12
1M: 8	**2M:** 16	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2025)

**See preface for how Avg Weightage and Range are calculated.*

Q3.26.1 MCQ 2026 2M

Let the relevant bandwidth B of a digital communication system be 1 MHz and

$$kT = -174 \text{ dBm/Hz},$$

where k is Boltzmann's constant and T is the equivalent noise temperature of the receiver.

The received signal power through an additive Gaussian channel is $S = -80$ dBm.

Which of the following statements are true about the Shannon capacity C of the channel?

A. $C = B$ **B.** $C = 2B$

C. $C > 3B$ **D.** $C < B$

Q3.25.1 NAT 2025 2M

X and Y are Bernoulli random variables taking values in $\{0, 1\}$. The joint probability mass function is given by

$$P(X = 0, Y = 0) = 0.06$$

$$P(X = 0, Y = 1) = 0.14$$

$$P(X = 1, Y = 0) = 0.24$$

$$P(X = 1, Y = 1) = 0.56$$

The mutual information $I(X;Y)$ is _____ (rounded off to two decimal places).

Q3.25.2 MSQ 2025 2M

The random variable X takes values in

$$\{-1, 0, 1\}$$

with probabilities

$$P(X = -1) = P(X = 1) = \alpha$$

and

$$P(X = 0) = 1 - 2\alpha, \qquad 0 < \alpha < \frac{1}{2}.$$

Let $g(\alpha)$ denote the entropy of X (in bits), parameterized by α.

Which of the following statements is/are TRUE?

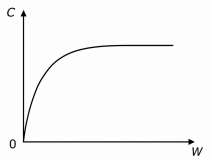
A. $g(0.4) > g(0.3)$ **B.** $g(0.3) > g(0.4)$

C. $g(0.3) > g(0.25)$ **D.** $g(0.25) > g(0.3)$

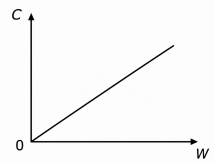
Q3.25.3 MCQ 2025 1M

Consider an additive white Gaussian noise (AWGN) channel with bandwidth W and noise power spectral density $\frac{N_0}{2}$. Let P_{av} denote the average transmit power constraint. Which one of the following plots illustrates the dependence of the channel capacity C on the bandwidth W (keeping P_{av} and N_0 fixed)?

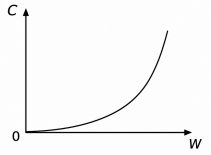
A.



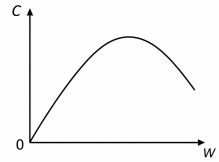
B.



C.



D.



Q3.24.1 NAT 2024 1M

A source transmits symbols from an alphabet of size 16. The value of maximum achievable entropy (in bits) is _____.

Q3.23.1 NAT 2023 2M

The frequency of occurrence of 8 symbols $(a - h)$ is given below:

Symbols	a	b	c	d	e	f	g	h
Frequency of occurrence	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{16}$	$\frac{1}{32}$	$\frac{1}{64}$	$\frac{1}{128}$	$\frac{1}{128}$

Fig. Q3.23.1

A symbol is chosen and it is determined by asking a series of "yes/no" questions which are truthfully answered.

The average number of questions when asked in the most efficient sequence, to determine the chosen symbol, is _____(rounded off to two decimal places).

Q3.22.1 NAT 2022 2M

The transition diagram of a discrete memoryless channel with three input symbols and three output symbols is shown below.

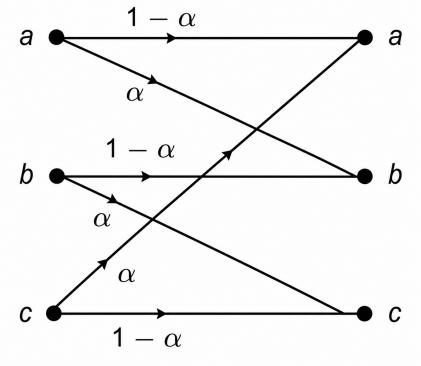


Fig. Q3.22.1

The transition probabilities are as marked.The parameter α lies in the interval $[0.25, 1]$.
The value of α for which the capacity of this channel is maximized is _____(rounded off to two decimal places).

Q3.22.2 MSQ | 2022 | 1M

Let $H(X)$ denote the entropy of a discrete random variable X taking K possible distinct real values.Which of the following statements is/are necessarily true?

A. $H(X) \leq \log_2 K$ bits
B. $H(X) \leq H(2X)$
C. $H(X) \leq H(X^2)$
D. $H(X) \leq H(2^X)$

Q3.17.1 MCQ | 2017 | 2M

Consider a binary memoryless channel characterized by the transition probability diagram shown in the figure.The channel is

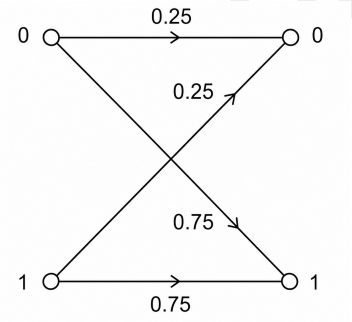
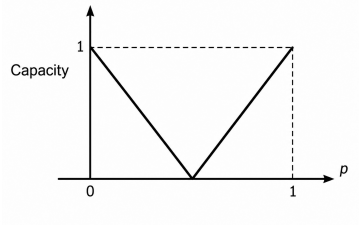


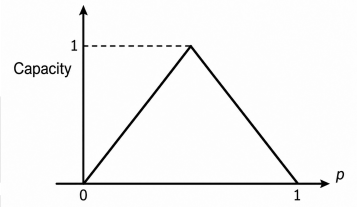
Fig. Q3.17.1

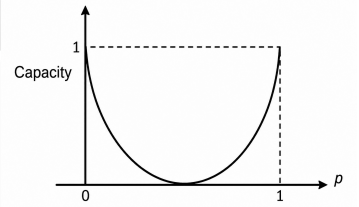
A. Lossless
B. Noiseless
C. Useless
D. Deterministic

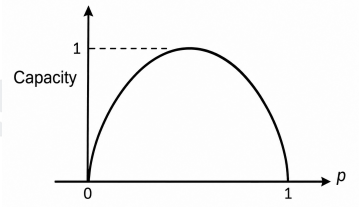
Q3.17.2 MCQ | 2017 | 1M

Which one of the following graphs shows the Shannon capacity (channel capacity) in bits of a memoryless binary symmetric channel with crossover probability p?

A.


B.


C.


D.


Q3.17.3 NAT | 2017 | 1M

Let (X_1, X_2) be independent random variables. X_1 has mean 0 and variance 1, while X_2 has mean 1 and variance 4.
The mutual information $I(X_1; X_2)$ between X_1 and X_2 in bits is _____.

Q3.16.1 NAT | 2016 | 1M

An analog baseband signal, bandlimited to 100 Hz, is sampled at the Nyquist rate. The samples are quantized into four message symbols that occur independently with probabilities $p_1 = p_4 = 0.125, p_2 = p_3$. The information rate (bits/sec) of the message source is _____.

Q3.16.2 NAT 2016 2M

Consider a discrete memoryless source with alphabet

$$S = \{s_0, s_1, s_2, s_3, \ldots\}$$

and respective probabilities of occurrence

$$P = \left\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \ldots\right\}.$$

The entropy of the source (in bits) is _____.

Q3.16.3 NAT 2016 2M

A binary communication system makes use of the symbols "zero" and "one". There are channel errors. Consider the following events:

x_0 : "zero" is transmitted

x_1 : "one" is transmitted

y_0 : "zero" is received

y_1 : "one" is received

The following probabilities are given:

$$P(x_0) = \frac{1}{2}, \qquad P(y_0|x_0) = \frac{3}{4}, \qquad P(y_0|x_1) = \frac{1}{2}.$$

The information in bits that you obtain when you learn which symbol has been received (while you know that a "zero" has been transmitted) is _____.

Q3.14.1 MCQ 2014 2M

The capacity of a band-limited additive white Gaussian noise (AWGN) channel is given by

$$C = W \log_2\left(1 + \frac{P}{\sigma^2 W}\right) \quad \text{bits/sec},$$

where W is the channel bandwidth, P is the average received power, and σ^2 is the one-sided power spectral density of the AWGN.
For a fixed

$$\frac{P}{\sigma^2} = 1000,$$

the channel capacity (in kbps) with infinite bandwidth ($W \to \infty$) is approximately

A. 1.44 **B.** 1.08
C. 0.72 **D.** 0.36

Q3.14.2 NAT 2014 2M

A fair coin is tossed repeatedly until a "Head" appears for the first time. Let L be the number of tosses required to obtain the first Head. The entropy $H(L)$ in bits is _____.

Q3.14.3 NAT 2014 2M

Consider the Z-channel given in the figure. The input is 0 or 1 with equal probability.If the output is 0, the probability that the input is also 0 equals _____

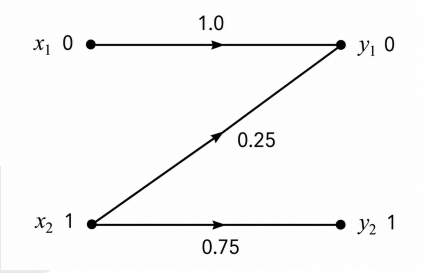


Fig. Q3.14.3

Q3.14.4 NAT 2014 2M

A binary random variable X takes the value of 1 with probability 1/3. X is input to a cascade of 2 independent identical binary symmetric channels (BSCs) each with crossover probability 1/2. The output of BSCs are the random variables Y_1 and Y_2 as shown in the figure.The value of $H(Y_1) + H(Y_2)$ in bits is _____

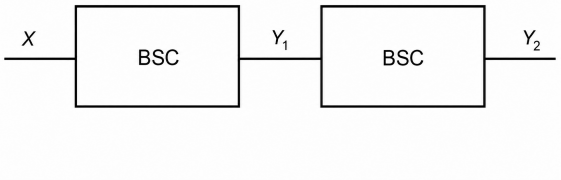


Fig. Q3.14.4

Q3.14.5 NAT 2014 1M

The capacity of a Binary Symmetric Channel (BSC) with cross-over probability 0.5 is _____.

Q3.12.1 MCQ 2012 1M

A source alphabet consists of N symbols with the probability of the first two symbols being the same. A source encoder increases the probability of the first symbol by a small amount ε and decreases that of the second by ε. After encoding, the entropy of the source

A. increases
B. remains the same
C. increases only if $N = 2$
D. decreases

Q3.09.1 MCQ 2009 2M

A communication channel with AWGN operating at a signal-to-noise ratio

$$\text{SNR} \gg 1$$

and bandwidth B has capacity C_1. If the SNR is doubled while keeping the bandwidth constant, the resulting capacity C_2 is given by

A. $C_2 \approx 2C_1$
B. $C_2 \approx C_1 + B$
C. $C_2 \approx C_1 + 2B$
D. $C_2 \approx C_1 + 0.3B$

Q3.08.1 MCQ 2008 2M

A memoryless source emits n symbols, each with probability p. The entropy of the source as a function of n is

A. increases as $\log n$
B. decreases as $\log(1/n)$
C. increases as n
D. increases as $n \log n$

Q3.90.1 MCQ 1990 2M

An image uses 512×512 picture elements. Each picture element can take any one of the 8 distinguishable intensity levels.

The maximum entropy in the above image will be

A. 2097152 bits
B. 786432 bits
C. 648 bits
D. 144 bits

Q3.89.1 MCQ 1989 2M

A source produces 4 symbols with probabilities

$$\frac{1}{2}, \quad \frac{1}{4}, \quad \frac{1}{8}, \quad \frac{1}{8}.$$

For this source, a practical coding scheme has an average codeword length of

$$L = 2 \text{ bits/symbol}.$$

The efficiency of the code is

A. 1
B. $\frac{7}{8}$
C. 0.5
D. 0.25

Answer Key

Q3.26.1 C	**Q3.25.1** 0	**Q3.25.2** B,C
Q3.25.3 A	**Q3.24.1** 4	**Q3.23.1** 1.98
Q3.22.1 1	**Q3.22.2** A,B,D	**Q3.17.1** C
Q3.17.2 C	**Q3.17.3** 0	**Q3.16.1** 360 to 363
Q3.16.2 1.8 to 2.2	**Q3.16.3** 0.8 to 0.82	**Q3.14.1** A
Q3.14.2 1.99 to 2.01	**Q3.14.3** 0.79 to 0.81	**Q3.14.4** 1.9 to 2.1
Q3.14.5 0	**Q3.12.1** D	**Q3.09.1** B
Q3.08.1 A	**Q3.90.1** B	**Q3.89.1** B

UNIT

IV

Analog Communications

Topics

1. Amplitude Modulation and Demodulation
2. Spectra of AM Signals
3. Angle Modulation - FM and PM
4. Demodulation of FM and PM
5. Spectra of FM Signals
6. Superheterodyne Receivers

PrepFusion

Unit Snapshot*

Stream: EC	**Questions:** 84	**MCQ:** 63	**MSQ:** 3	**NAT:** 18
1M: 36	**2M:** 48	**Descriptive:** 0	**Avg Weightage** ~3M	**Range:** 1M(2024)–5M(2021)

**See preface for how Avg Weightage and Range are calculated.*

Q4.26.1 MSQ 2026 2M

Consider a real, narrowband signal

$$x(t) = A(t)\cos(2\pi f_c t + \theta(t))$$

where the maximum frequency components of $A(t)$ and $\theta(t)$ are f_m and $f_c = 1000 f_m$, respectively.
Which of the following statements is/are correct for

$$-\infty < t < \infty\,?$$

A. $x(t)$ represents a PSK modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
B. $x(t)$ represents an amplitude modulated signal for suitable choices of $A(t)$ and $\theta(t)$.
C. $x(t)$ represents a band-limited Gaussian noise process.
D. $x(t)$ never represents a narrowband FM signal.

Q4.25.1 MSQ 2025 1M

Consider a message signal $m(t)$ which is bandlimited to [-W,W], where W is in Hz.Consider the following two modulation schemes: **Double Sideband Suppressed Carrier (DSB-SC):**

$$f_{\text{DSB}}(t) = A_c\, m(t)\cos(2\pi f_c t)$$

Amplitude Modulation (AM):

$$f_{\text{AM}}(t) = A_c\big(1 + \mu m(t)\big)\cos(2\pi f_c t)$$

where A_c and f_c are the amplitude and frequency of the carrier, respectively, and μ is the modulation index.Consider the following statements:

(i) An envelope detector can be used for demodulation in the DSB-SC scheme if $m(t) > 0$ for all t.

(ii) An envelope detector can be used for demodulation in the AM scheme only if $m(t) > 0$ for all t.

Which of the following options is/are correct?

A. (i) is TRUE **B.** (i) is FALSE
C. (ii) is TRUE **D.** (ii) is FALSE

Q4.25.2 MCQ 2025 1M

Consider a frequency-modulated (FM) signal

$$f(t) = A_c\cos\left(2\pi f_c t + 3\sin(2\pi f_1 t) + 4\sin(6\pi f_1 t)\right),$$

where A_c and f_c are, respectively, the amplitude and frequency (in Hz) of the carrier waveform. The frequency f_1 is in Hz, and assume that

$$f_c > 100 f_1.$$

The peak frequency deviation of the FM signal in Hz is _____

A. $15 f_1$ **B.** $12 f_1$
C. $4 f_1$ **D.** $2 f_1$

Q4.24.1 NAT 2024 1M

An amplitude modulator has output (in Volts)

$$s(t) = A\cos(400\pi t) + B\cos(360\pi t) + B\cos(440\pi t).$$

The carrier power normalized to a $1\,\Omega$ resistance is 50 Watts. The ratio of the total sideband power to the total power is $\frac{1}{9}$. The value of B (in Volts), rounded off to two decimal places, is _____

Q4.23.1 MCQ 2023 2M

Let a frequency modulated (FM) signal be

$$x(t) = A\cos\left(\omega_c t + k_f \int_{-\infty}^{t} m(\lambda)\, d\lambda\right),$$

where $m(t)$ is a message signal of bandwidth W. Let B_T denote the FM bandwidth.The signal is passed through a nonlinear system whose output is

$$y(t) = 2x(t) + 5x^2(t).$$

The minimum value of ω_c required to recover $x(t)$ from $y(t)$ is

A. $B_T + W$ **B.** $\frac{3}{2}B_T$
C. $2B_T + W$ **D.** $\frac{5}{2}B_T$

Q4.22.1 MSQ 2022 2M

Consider an FM broadcast that employs a pre-emphasis filter with frequency response

$$H_{pe}(\omega) = 1 + j\frac{\omega}{\omega_0}$$

where

$$\omega_0 = 10^4 \text{ rad/s}.$$

For the RC network shown below to act as the corresponding de-emphasis filter, the appropriate pair(s) of (R, C) values is/are:

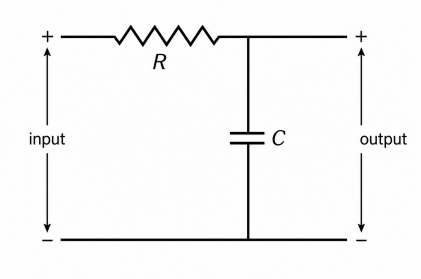


Fig. Q4.22.1

A. $R = 1\,\text{k}\Omega,\ C = 0.1\,\mu\text{F}$ **B.** $R = 2\,\text{k}\Omega,\ C = 1\,\mu\text{F}$
C. $R = 1\,\text{k}\Omega,\ C = 2\,\mu\text{F}$ **D.** $R = 2\,\text{k}\Omega,\ C = 0.5\,\mu\text{F}$

Q4.21.1 NAT 2021 2M

A superheterodyne receiver is tuned to 600 kHz. If the local oscillator feeds a 1000 kHz signal to the mixer, the image frequency (in integer) is _____ kHz.

Q4.21.2 NAT 2021 2M

A sinusoidal message signal having root mean square value of 4 V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2 rad/V. If the carrier signal is

$$c(t) = 2\cos(2\pi \times 10^6 t),$$

the maximum instantaneous frequency of the phase modulated signal (rounded off to one decimal place) is _____ Hz.

Q4.21.3 NAT 2021 1M

Consider a carrier signal which is amplitude modulated by a single-tone sinusoidal message signal with a modulation index of 50%.If the carrier and one of the sidebands are suppressed in the modulated signal, the percentage of power saved (rounded off to one decimal place) is _____.

Q4.20.1 NAT 2020 2M

The phase modulated and frequency modulated signals corresponding to the message signal $m(t)$ are given by

$$S_{PM}(t) = \cos\big(1000\pi t + K_p m(t)\big)$$

and

$$S_{FM}(t) = \cos\left(1000\pi t + K_f \int_{-\infty}^{t} m(\tau)\, d\tau\right),$$

where K_p is the phase deviation constant (rad/V) and K_f is the frequency deviation constant (rad/s/V). If the highest instantaneous frequencies of $S_{PM}(t)$ and $S_{FM}(t)$are equal, determine

$$\frac{K_p}{K_f}.$$

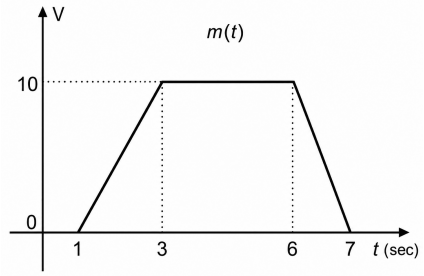


Fig. Q4.20.1

Q4.20.2 MCQ 2020 2M

For the modulated signal $x(t) = m(t)\cos(2\pi f_c t)$, the message signal is $m(t) = 4\cos(1000\pi t)$, and the carrier frequency is $f_c = 1$ MHz.The signal $x(t)$ is applied to a coherent demodulator consisting of a multiplier and an ideal low-pass filter (LPF). The local oscillator signal is $\cos\big(2\pi(f_c + 40)t\big)$, and the LPF has cut-off frequencies ±510 Hz. Determine the output $y(t)$ of the demodulator.

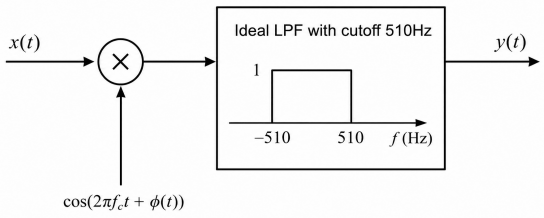


Fig. Q4.20.2

Q4.19.1 NAT 2019 1M

The baseband signal $m(t)$ shown in the figure is phase-modulated to generate the PM signal

$$\phi(t) = 2\cos\big(2\pi f_c t + k\, m(t)\big).$$

The time axis in the figure is in milliseconds. If the carrier frequency is $f_c = 50$ kHz and $k = 10\pi$,then the ratio of the minimum instantaneous frequency (in kHz) to the maximum instantaneous frequency (in kHz) is _____.(rounded off to two decimal places).

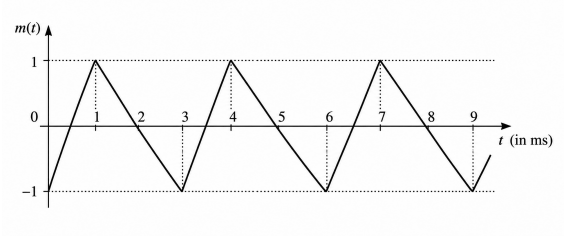


Fig. Q4.19.1

Q4.18.1 MCQ 2018 2M

Let

$$c(t) = A_c \cos(2\pi f_c t)$$

and

$$m(t) = \cos(2\pi f_m t).$$

It is given that

$$f_c \gg 5f_m.$$

The signal

$$v_i(t) = c(t) + m(t)$$

is applied to the input of a non-linear device whose output is

$$v_o(t) = a\, v_i(t) + b\, v_i^2(t),$$

where a and b are positive constants.The output of the non-linear device is passed through an ideal band-pass filter (BPF) having center frequency f_c and bandwidth $3f_m$, producing an amplitude-modulated (AM) wave. If the sideband power of the AM wave is half of the carrier power, then the ratio

$$\frac{a}{b}$$

is

A. 0.25 **B.** 0.5
C. 1 **D.** 2

Q4.18.2 NAT 2018 1M

Consider the following amplitude modulated signal:

$$s(t) = \cos(2000\pi t) + 4\cos(2400\pi t) + \cos(2800\pi t).$$

The ratio (accurate to three decimal places) of the power of the message signal to the power of the carrier signal is _____.

Q4.17.1 NAT 2017 2M

A modulating signal is given by

$$x(t) = 5\sin\left(4\pi \times 10^3 t - 10\pi\cos(2\pi \times 10^3 t)\right)\ \text{V}$$

and is fed to a phase modulator with phase deviation constant $k_p = 5$ rad/V. If the carrier frequency is $f_c = 20$ kHz, the instantaneous frequency (in kHz) at $t = 0.5$ ms is _____

Q4.17.2 NAT 2017 2M

The un-modulated carrier power in an AM transmitter is 5 kW. This carrier is modulated by a sinusoidal modulating signal. The maximum percentage modulation is 50%. If it is reduced to 40%, then the maximum un-modulated carrier power (in kW) that can be used without overloading the transmitter is _____

Q4.16.1 NAT 2016 1M

A superheterodyne receiver operates in the frequency range of 58 MHz – 68 MHz.The intermediate frequency f_{IF} and local oscillator frequency f_{LO} are chosen such that

$$f_{IF} \le f_{LO}.$$

It is required that the image frequencies fall outside the 58 MHz – 68 MHz band. The minimum required f_{IF} (in MHz) is _____

Q4.16.2 NAT 2016 1M

For a superheterodyne receiver, the intermediate frequency is $f_{IF} = 15$ MHz and the local oscillator frequency is $f_{LO} = 3.5$ GHz. If the frequency of the received signal is greater than the local oscillator frequency, then the image frequency (in MHz) is _____

Q4.16.3 NAT 2016 1M

The amplitude of a sinusoidal carrier is modulated by a single-tone sinusoid to obtain the amplitude modulated signal

$$s(t) = 5\cos(1600\pi t) + 20\cos(1800\pi t) + 5\cos(2000\pi t).$$

The value of the modulation index is _____

Q4.16.4 MCQ 2016 1M

A frequency synthesizer consists of a Phase Locked Loop (PLL) and a divide-by-N counter. The divider provides outputs corresponding to

$$\div 2, \quad \div 4, \quad \div 8, \quad \div 16.$$

The synthesizer is excited with an input signal of frequency

$$f_{in} = 5 \text{ kHz}.$$

The free-running frequency of the VCO is 20 kHz. The commutator switch selects the divider outputs repeatedly in the order

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4,$$

corresponding to $N = 2, 4, 8, 16$. The corresponding frequencies synthesized are

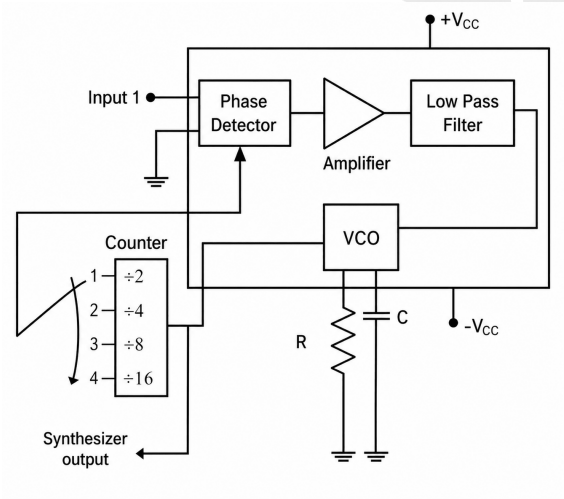


Fig. Q4.16.4

A. 10 kHz, 20 kHz, 40 kHz, 80 kHz
B. 20 kHz, 40 kHz, 80 kHz, 160 kHz
C. 80 kHz, 40 kHz, 20 kHz, 10 kHz
D. 160 kHz, 80 kHz, 40 kHz, 20 kHz

Q4.15.1 MCQ 2015 2M

The complex envelope of the bandpass signal

$$x(t) = -\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)\sin\left(\pi t - \frac{\pi}{4}\right)$$

centered about $f_c = \frac{1}{2}$ Hzis _____

A.
$$\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}$$

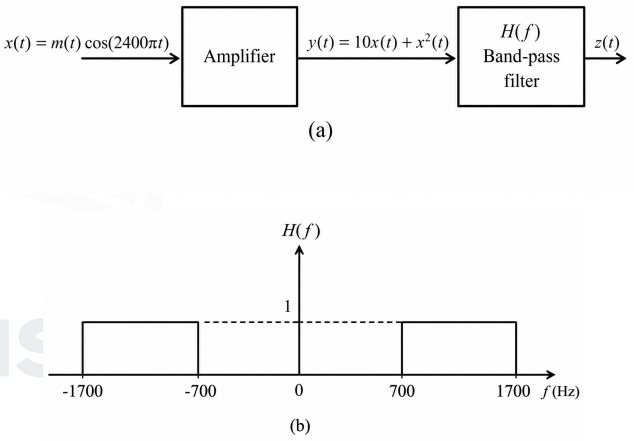
B.
$$\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{-j\frac{\pi}{4}}$$

C.
$$\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{j\frac{\pi}{4}}$$

D.
$$\sqrt{2}\left(\frac{\sin\left(\frac{\pi t}{5}\right)}{\frac{\pi t}{5}}\right)e^{-j\frac{\pi}{4}}$$

Q4.15.2 NAT 2015 2M

In the system shown in Figure (a), $m(t)$ is a low-pass signal with bandwidth W Hz. The frequency response of the band-pass filter $H(f)$ is shown in Figure (b). If it is desired that the output signal $z(t) = 10x(t)$, the maximum value of W (in Hz) should be strictly less than _____



Q4.15.3 MCQ 2015 1M

A message signal $m(t) = A_m \sin(2\pi f_m t)$ is used to modulate the phase of a carrier $A_c \cos(2\pi f_c t)$ to obtain the phase-modulated signal

$$y(t) = A_c \cos(2\pi f_c t + m(t)).$$

The bandwidth of $y(t)$ _____

A. depends on A_m but not on f_m
B. depends on f_m but not on A_m
C. depends on both A_m and f_m
D. does not depend on A_m or f_m

Q4.15.4 MCQ 2015 1M

Consider the signal

$$s(t) = m(t)\cos(2\pi f_c t) + \hat{m}(t)\sin(2\pi f_c t)$$

where $\hat{m}(t)$ denotes the Hilbert transform of $m(t)$, and the bandwidth of $m(t)$ is very small compared to f_c. The signal $s(t)$ is a _____

A. high-pass signal
B. low-pass signal
C. band-pass signal
D. double sideband suppressed carrier signal

Q4.14.1 NAT 2014 2M

If the figure, $M(f)$ is the Fourier transform of the message signal $m(t)$ where $A = 100$ Hz and $B = 40$ Hz. Given $v(t) = \cos(2\pi f_c t)$ and $w(t) = \cos\big(2\pi(f_c + A)t\big)$, where $f_c > A$. The cutoff frequencies of both the filters are f_c.The bandwidth of the signal at the output of the modulator (in Hz) is _____

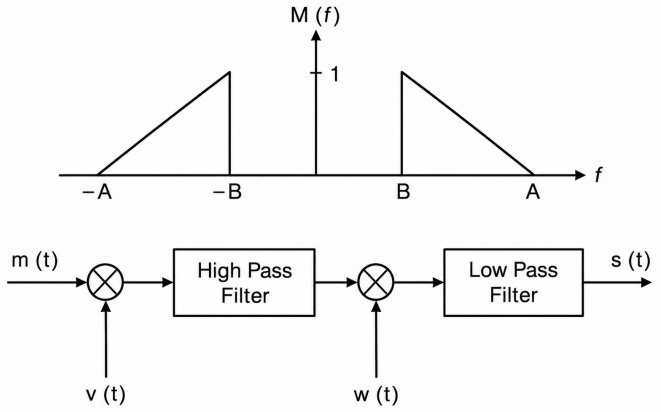


Fig. Q4.14.1

Q4.14.2 NAT 2014 1M

Consider sinusoidal modulation in an AM system. Assuming no overmodulation, the modulation index (μ) when the maximum and minimum values of the envelope, respectively, are 3 V and 1 V, is _____

Q4.14.3 MCQ 2014 1M

Consider a FM signal

$$f(t) = \cos[2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t)]\,.$$

The maximum deviation of the instantaneous frequency from the carrier frequency f_c is

A. $\beta_1 f_1 + \beta_2 f_2$
B. $\beta_1 f_2 + \beta_2 f_1$
C. $\beta_1 + \beta_2$
D. $f_1 + f_2$

Q4.14.4 NAT 2014 1M

In a double side-band (DSB) full carrier AM transmission system, if the modulation index is doubled, then the ratio of total sideband power to the carrier power increases by a factor of _____

Q4.12.1 MCQ 2012 2M

The signal $m(t)$ as shown is applied both to a phase modulator (with k_p as the phase constant) and a frequency modulator (with k_f as the frequency constant) having the same carrier frequency.

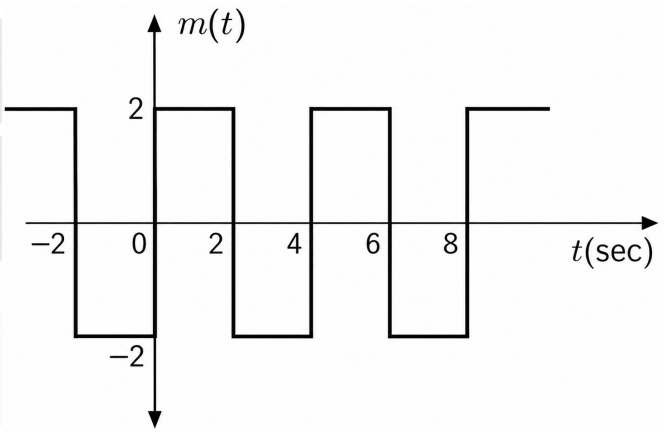


Fig. Q4.12.1

The ratio k_p/k_f (in rad/Hz) for the same maximum phase deviation is

A. 8π
B. 4π
C. 2π
D. π

Q4.11.1 MCQ 2011 2M

A message signal $m(t) = \cos 2000\pi t + 4\cos 4000\pi t$ modulates the carrier $c(t) = \cos 2\pi f_c t$ where $f_c = 1$MHz to produce an AM signal. For demodulating the generated AM signal using an envelope detector, the time constant RC of the detector circuit should satisfy

A. 0.5 ms $< RC <$ 1 ms
B. 1 μs $\ll RC <$ 0.5 ms
C. $RC \ll \mu$s
D. $RC \gg$ 0.5 ms

Q4.11.2 MCQ 2011 1M

The Column-I lists the attributes and the Column-II lists the modulation systems. Match the attribute to the modulation system that best meets it.

	Column-I		Column-II
P	Power efficient transmission of signals	1	Conventional AM
Q	Most bandwidth efficient transmission of voice signals	2	FM
R	Simplest receiver structure	3	VSB
S	Bandwidth efficient transmission of signals with Significant dc component	4	SSB-SC

Fig. Q4.11.2

A. P-4;Q-2;R-1;S-3 **B.** P-2;Q-4;R-1;S-3
C. P-3;Q-2;R-1;S-4 **D.** P-2;Q-4;R-3;S-1

Q4.10.1 MCQ 2010 1M

Consider an angle modulated signal

$x(t) = 6\cos\left[2\pi \times 10^6 t + 2\sin(8000\pi t) + 4\cos(8000\pi t)\right]$ V

The average power of $x(t)$ is

A. 10 W **B.** 18 W
C. 20 W **D.** 28 W

Q4.10.2 MCQ 2010 1M

Suppose that the modulating signal is $m(t) = 2\cos(2\pi f_m t)$ and the carrier signal is $X_c(t) = A_c\cos(2\pi f_c t)$, which one of the following is a conventional AM signal without over-modulation

A. $x(t) = A_c m(t)\cos(2\pi f_c t)$
B. $x(t) = A_c\,[1 + m(t)]\cos(2\pi f_c t)$
C. $x(t) = A_c\cos(2\pi f_c t) + \frac{A_c}{4}m(t)\cos(2\pi f_c t)$
D. $x(t) = A_c\cos(2\pi f_c t)\cos(2\pi f_c t) + A_c\sin(2\pi f_m t)\sin(2\pi f_c t)$

Q4.09.1 MCQ 2009 2M

A message signal given by

$$m(t) = \left(\frac{1}{2}\right)\cos\omega_1 t - \left(\frac{1}{2}\right)\sin\omega_2 t$$

is amplitude-modulated with a carrier of frequency ω_c to generate

$$s(t) = [1 + m(t)]\cos\omega_c t$$

What is the power efficiency achieved by this modulation scheme?

A. 8.33% **B.** 11.11%
C. 20% **D.** 25%

Q4.09.2 MCQ 2009 1M

For a message signal $m(t) = \cos(2\pi f_m t)$ and carrier of frequency f_c, which of the following represents a single side-band (SSB) signal?

A. $\cos(2\pi f_m t)\cos(2\pi f_c t)$
B. $\cos(2\pi f_c t)$
C. $\cos[2\pi(f_c + f_m)t]$
D. $[1 + \cos(2\pi f_m t)]\cos(2\pi f_c t)$

Q4.08.1 MCQ 2008 2M

The signal

$$\cos\omega_c t + 0.5\cos\omega_m t\,\sin\omega_c t$$

is

A. FM only **B.** AM only
C. both AM and FM **D.** neither AM nor FM

Q4.08.2 MCQ 2008 2M

Consider the frequency modulated signal

$$10\cos\left[2\pi \times 10^5 t + 5\sin(2\pi \times 1500t) + 7.5\sin(2\pi \times 1000t)\right]$$

with carrier frequency of 10^5 Hz. The modulation index is

A. 12.5 **B.** 10
C. 7.5 **D.** 5

Q4.08.3 MCQ 2008 1M

Consider the amplitude modulated (AM) signal

$$A_c\cos\omega_c t + 2\cos\omega_m t\cos\omega_c t.$$

For demodulating the signal using envelope detector, the minimum value of A_c should be

A. 2 **B.** 1
C. 0.5 **D.** 0

Q4.07.1 MCQ 2007 2M

In the following scheme, if the spectrum M(f) of m(t) is as shown, then the spectrum Y(f) of y(t) will be

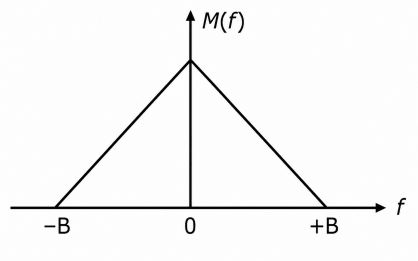


Fig. Q4.07.1

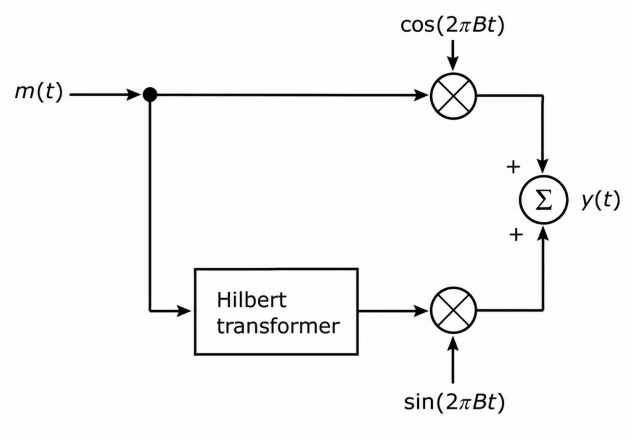
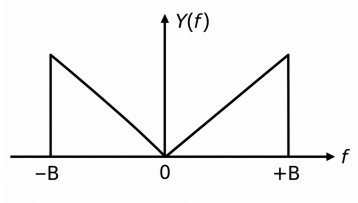
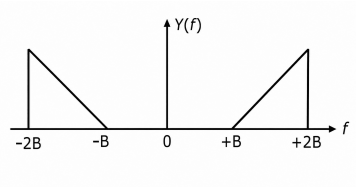
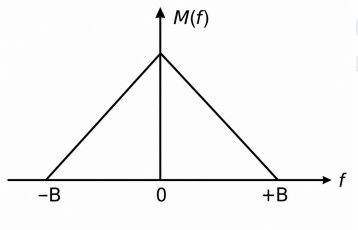
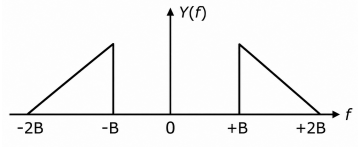


Fig. Q4.07.1

A.


B.


C.


D.


Q4.06.1 MCQ 2006 2M

Consider the following Amplitude Modulated (AM) signal, where $f_m < B$:

$$X_{AM}(t) = 10\,(1 + 0.5 \sin 2\pi f_m t) \cos 2\pi f_c t.$$

The AM signal gets added to a noise with Power Spectral Density $S_n(f)$ given in the figure below. The ratio of average sideband power to mean noise power would be

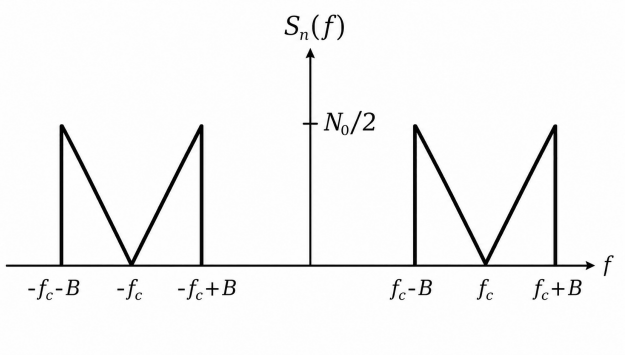


Fig. Q4.06.1

A. $\frac{25}{8N_0B}$ **B.** $\frac{25}{4N_0B}$

C. $\frac{25}{2N_0B}$ **D.** $\frac{25}{N_0B}$

Q4.06.2 MCQ 2006 2M

Consider the following Amplitude Modulated (AM) signal, where $f_m < B$:

$$X_{AM}(t) = 10\,(1 + 0.5 \sin 2\pi f_m t) \cos 2\pi f_c t.$$

The average side-band power for the AM signal given above is

A. 25 **B.** 12.5

C. 6.25 **D.** 3.125

Q4.06.3 MCQ 2006 2M

Let $g(t) = p(t) * p(t)$ where $*$ denotes convolution and $p(t) = u(t) - u(t-1)$ with $u(t)$ being the unit step function.An Amplitude Modulated signal is given as

$$X_{AM}(t) = 100\,(p(t) + 0.5g(t)) \cos \omega_c t$$

in the interval $0 \le t \le 1$. One set of possible values of modulating signal and modulation index would be

A. t, 0.5 **B.** t, 1.0

C. t, 2.0 **D.** t^2, 0.5

Q4.06.4 MCQ 2006 2M

The diagonal clipping in Amplitude Demodulation (using envelope detector) can be avoided if RC time-constant of the envelope detector satisfies the following condition, (here W is message bandwidth and ω_c is carrier frequency both in (rad/sec))

A. $RC < \frac{1}{W}$ **B.** $RC > \frac{1}{W}$

C. $RC < \frac{1}{\omega_c}$ **D.** $RC > \frac{1}{\omega_c}$

Q4.06.5 MCQ 2006 2M

A message signal with bandwidth 10 kHz is Lower-Side Band SSB modulated with carrier frequency $f_{c1} = 10^6$ Hz. The resulting signal is then passed through a Narrow-Band Frequency Modulator with carrier frequency $f_{c2} = 10^9$ Hz. The bandwidth of the output would be

A. 4×10^4 Hz **B.** 2×10^6 Hz
C. 2×10^9 Hz **D.** 2×10^{10} Hz

Q4.05.1 MCQ 2005 2M

A carrier is phase modulated (PM) with frequency deviation of 10 kHz by a single tone frequency of 1 kHz. If the single tone frequency is increased to 2 kHz, assuming that phase deviation remains unchanged, the bandwidth of the PM signal is

A. 21 kHz **B.** 22 kHz
C. 42 kHz **D.** 44 kHz

Q4.05.2 MCQ 2005 2M

A device with input $x(t)$ and output $y(t)$ is characterized by:

$$y(t) = x^2(t).$$

An FM signal with frequency deviation of 90 kHz and modulating signal bandwidth of 5 kHz is applied to this device. The bandwidth of the output signal is

A. 370 kHz **B.** 190 kHz
C. 380 kHz **D.** 95 kHz

Q4.05.3 MCQ 2005 1M

Which of the following analog modulation scheme requires the minimum transmitted power and minimum channel bandwidth ?

A. VSB **B.** $DSB - SC$
C. SSB **D.** AM

Q4.05.4 MCQ 2005 1M

Find the correct match between group 1 and group 2.

Group 1	Group 2
P - $\{1 + km(t)\}A\sin(\omega_c t)$	W – Phase modulation
Q - $km(t)\,A\sin(\omega_c t)$	X – Frequency modulation
R - $A\sin\{\omega_c t + km(t)\}$	Y – Amplitude modulation
S - $A\sin\left(\omega_c t + k\int_{-\infty}^{t} m(\tau)\,d\tau\right)$	Z – DSB-SC modulation

Fig. Q4.05.4

A. P-Z,Q-Y,R-X,S-W **B.** P-W,Q-X,R-Y,S-Z
C. P-X;Q-W;R-Z;S-Y **D.** P-Y;Q-Z;R-W;S-X

Q4.04.1 MCQ 2004 2M

Consider a system shown in the figure. Let $X(f)$ and $Y(f)$ denote the Fourier transforms of $x(t)$ and $y(t)$ respectively. The ideal HPF has the cutoff frequency 10 kHz.

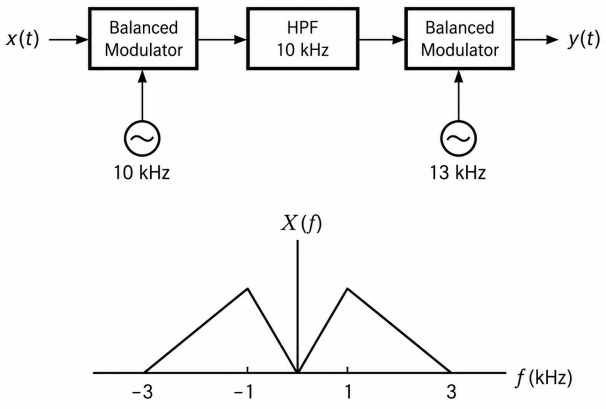


Fig. Q4.04.1

The positive frequencies where $Y(f)$ has spectral peaks are

A. 1 kHz and 24 kHz **B.** 2 kHz and 24 kHz
C. 1 kHz and 14 kHz **D.** 2 kHz and 14 kHz

Q4.04.2 MCQ 2004 2M

Two sinusoidal signals of same amplitude and frequencies 10 kHz and 10.1 kHz are added together. The combined signal is given to an ideal frequency detector. The output of the detector is

A. 0.1 kHz sinusoid **B.** 20.1 kHz sinusoid
C. a linear function of time **D.** a constant

Q4.04.3 MCQ 2004 2M

A 100 MHz carrier of 1 V amplitude and a 1 MHz modulating signal of 1 V amplitude are fed to a balanced modulator. The output of the modulator is passed through an ideal high-pass filter with cut-off frequency of 100 MHz. The output of the filter is added with 100 MHz signal of 1 V amplitude and 90° phase shift as shown in the figure. The envelope of the resultant signal is

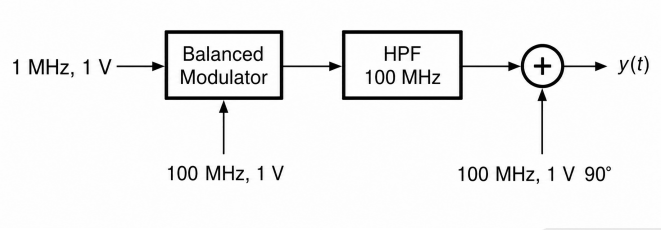


Fig. Q4.04.3

A. constant **B.** $\sqrt{1+\sin(2\pi\times10^6 t)}$

C. $\sqrt{\frac{5}{4}-\sin(2\pi\times10^6 t)}$ **D.** $\sqrt{\frac{5}{4}+\cos(2\pi\times10^6 t)}$

Q4.04.4 MCQ 2004 1M

An AM signal and a narrow-band FM signal with identical carriers, modulating signals and modulation indices of 0.1 are added together. The resultant signal can be closely approximated by

A. broadband FM **B.** SSB with carrier
C. DSB-SC **D.** SSB without carrier

Q4.04.5 MCQ 2004 1M

An AM signal is detected using an envelope detector. The carrier frequency and modulating signal frequency are 1 MHz and 2 kHz respectively. An appropriate value for the time constant of the envelope detector is

A. 500 μsec **B.** 20 μsec
C. 0.2 μsec **D.** 1 μsec

Q4.03.1 MCQ 2003 2M

A superheterodyne receiver is to operate in the frequency range 550 kHz–1650 kHz, with the intermediate frequency of 450 kHz. Let $R = C_{max}/C_{min}$ denote the required capacitance ratio of the local oscillator and I denote the image frequency (in kHz) of the incoming signal. If the receiver is tuned to 700 kHz, then

A. $R = 4.41,\ I = 1600$ **B.** $R = 2.10,\ I = 1150$
C. $R = 3.0,\ I = 1600$ **D.** $R = 9.0,\ I = 1150$

Q4.03.2 MCQ 2003 2M

Choose the correct one from among the alternative A,B,C,D after matching an item in Group 1 with most appropriate item in Group 2.

Group 1		Group 2	
P	Ring modulator	1	Clock recovery
Q	VCO	2	Demodulation of FM
R	Foster-Seely discriminator	3	Frequency conversion
S	Mixer	4	Summing the two inputs
		5	Generation of FM
		6	Generation of DSB-SC

Fig. Q4.03.2

A. P-1,Q-3,R-2,S-4 **B.** P-6,Q-5,R-2,S-3
C. P-6;Q-1;R-3;S-2 **D.** P-5;Q-6;R-1;S-3

Common Data Questions

The next 2 questions share the data below.

Answer the question 4.03.3 and question 4.03.4 using the following data.
Let $m(t) = \cos\left[(4\pi\times10^3)t\right]$ be the message signal and $c(t) = 5\cos\left[(2\pi\times10^6)t\right]$ be the carrier.

Q4.03.3 MCQ 2003 2M

$c(t)$ and $m(t)$ are used to generate an FM signal. If the peak frequency deviation of the generated FM signal is three times the transmission bandwidth of the AM signal, then the coefficient of the term $\cos\left[2\pi(1008\times10^3)t\right]$ in the FM signal (in terms of the Bessel coefficients) is

A. $5J_4(3)$ **B.** $\frac{5}{2}J_8(3)$

C. $\frac{5}{2}J_8(4)$ **D.** $5J_4(6)$

Q4.03.4 MCQ 2003 2M

$c(t)$ and $m(t)$ are used to generate an AM signal. The modulation index of the generated AM signal is 0.5. Then the quantity

$$\frac{\text{Total sideband power}}{\text{Carrier power}}$$

is

A. $\frac{1}{2}$ **B.** $\frac{1}{4}$

C. $\frac{1}{3}$ **D.** $\frac{1}{8}$

Q4.03.5 MCQ 2003 2M

A DSB-SC signal is to be generated with a carrier frequency $f_c = 1$ MHz using a non-linear device with the input-output characteristic

$$V_0 = a_0 v_i + a_1 v_i^3$$

where a_0 and a_1 are constants. The output of the non-linear device can be filtered by an appropriate band pass filter.Let

$$v_i = A_c' \cos(2\pi f_c' t) + m(t),$$

where $m(t)$ is the message signal. Then the value of f_c' (in MHz) is

A. 1.0 **B.** 0.333

C. 0.5 **D.** 3.0

Q4.03.6 MCQ 2003 1M

The input to a coherent detector is DSB-SC signal plus noise. The noise at the detector output is

A. the in-phase component **B.** the quadrature-component

C. zero **D.** the envelope

Q4.02.1 MCQ 2002 2M

In the figure,

$$m(t) = \frac{2\sin 2\pi t}{t}, \qquad s(t) = \cos 200\pi t$$

and

$$n(t) = \frac{\sin 199\pi t}{t}.$$

The output $y(t)$ will be

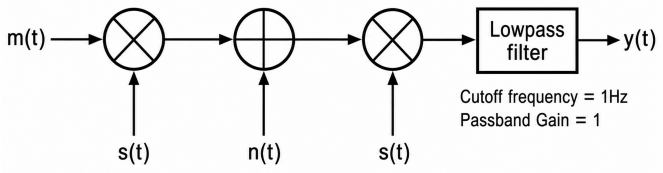


Fig. Q4.02.1

A. $\frac{\sin 2\pi t}{t}$

B. $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t}\cos 3\pi t$

C. $\frac{\sin 2\pi t}{t} + \frac{\sin 0.5\pi t}{t}\cos 1.5\pi t$

D. $\frac{\sin 2\pi t}{t} + \frac{\sin \pi t}{t}\cos 0.75\pi t$

Q4.02.2 MCQ 2002 2M

An angle modulated signal is given by

$$s(t) = \cos 2\pi \left(2 \times 10^6 t + 30\sin 150t + 40\cos 150t\right)$$

The maximum frequency and phase deviations of $s(t)$ are

A. 10.5 kHz, 140π rad **B.** 6 kHz, 80π rad

C. 10.5 kHz, 100π rad **D.** 7.5 kHz, 100π rad

Q4.02.3 MCQ 2002 1M

A 1 MHz sinusoidal carrier is amplitude modulated by a symmetrical square wave of period 100μsec. Which of the following frequencies will NOT be present in the modulated signal?

A. 990 kHz **B.** 1010 kHz

C. 1020 kHz **D.** 1030 kHz

Q4.00.1 MCQ 2000 2M

In an FM system, a carrier of 100MHz is modulated by a sinusoidal signal of 5kHz. The bandwidth by Carson's approximation is 1MHz.If $y(t) = (\text{modulated waveform})^3$, then by using Carson's approximation, the bandwidth of $y(t)$ around 300MHz and the spacing of spectral components are, respectively.

A. 3 MHz, 5 kHz **B.** 1 MHz, 15 kHz

C. 3 MHz, 15 kHz **D.** 1 MHz, 5 kHz

Q4.00.2 MCQ 2000 2M

A message $m(t)$ band limited to the frequency f_m has a power of P_m. The power of the output signal in the figure is

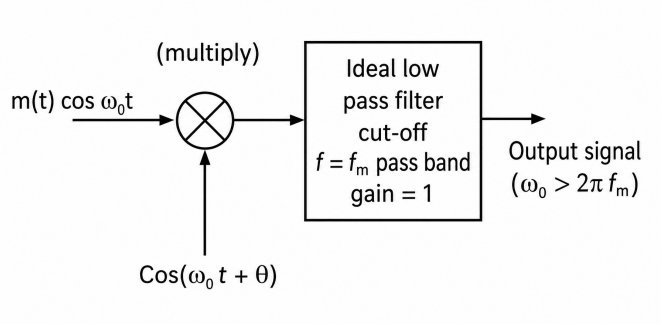


Fig. Q4.00.2

A. $\frac{P_m \cos\theta}{2}$ **B.** $\frac{P_m}{4}$

C. $\frac{P_m \sin^2\theta}{4}$ **D.** $\frac{P_m \cos^2\theta}{4}$

Q4.00.3 MCQ 2000 1M

The amplitude modulated wave form

$$s(t) = A_c\,[1 + K_a m(t)]\cos\omega_c t$$

is fed to an ideal envelope detector. The maximum magnitude of $K_a m(t)$ is greater than 1. Which of the following could be the detector output?

A. $A_c m(t)$ **B.** $A_c^2\,[1 + K_a m(t)]^2$

C. $|A_c\,[1 + K_a m(t)]|$ **D.** $A_c\,[1 + K_a m(t)]^2$

Q4.99.1 MCQ 1999 1M

A modulated signal is given by,

$$s(t) = m_1(t)\cos(2\pi f_c t) + m_2(t)\sin(2\pi f_c t),$$

where the base band signal $m_1(t)$ and $m_2(t)$ have bandwidths of 10 kHz and 15 kHz, respectively. The bandwidth of the modulated signal, in kHz, is

A. 10 kHz **B.** 15 kHz

C. 25 kHz **D.** 30 kHz

Q4.98.1 MCQ 1998 1M

A DSB-SC signal is generated using the carrier $\cos(\omega_c t+\theta)$ and modulating signal $x(t)$. The envelope of the DSB-SC signal is

A. $x(t)$ **B.** $|x(t)|$

C. only positive portion of $x(t)$ **D.** $x(t)\cos\theta$

Q4.98.2 MCQ 1998 1M

The image channel selectivity of super heterodyne receiver depends upon

A. IF amplifiers only

B. RF and IF amplifiers only

C. Pre selector, RF and IF amplifiers

D. Pre selector and RF amplifiers only

Q4.96.1 MCQ 1996 1M

The image channel rejection in a super heterodyne reciever comes from

A. IF stages only

B. RF stages only

C. detector and RF stages only

D. detector, RF and IF stages

Q4.96.2 MCQ 1996 2M

An FM signal with a modulation index 9 is applied to a frequency tripler. The modulation index in the output signal will be

A. 0 **B.** 3

C. 9 **D.** 27

Q4.95.1 MCQ 1995 1M

A PAM signal can be detected by using

A. an ADC **B.** an integrator

C. a band pass filter **D.** a high pass filter

Q4.95.2 MCQ 1995 1M

A PLL can be used to demodulate

A. PAM signals **B.** PCM signals

C. FM signals **D.** DSB-SC signals

Q4.95.3 MCQ 1995 1M

The image(second) channel selectivity of a super heterodyne communication receiver is determined by

A. antenna and pre selector

B. the pre selector and RF amplifier

C. the pre selector and IF amplifier

D. the RF and IF amplifier

Q4.94.1 MCQ 1994 2M

Match the following

List I	List II
a) SSB	1) Envelope detector
b) AM	2) Integrate and dump
c) BPSK	3) Hilbert transform
	4) Ratio detector
	5) PLL

Fig. Q4.94.1

Q4.94.2 MCQ 1994 1M

$v(t) = 5\left[\cos(10^6\pi t) - \sin(10^3\pi t) \times \sin(10^6\pi t)\right]$ represents

A. DSB suppressed carrier signal

B. AM signal

C. SSB upper sideband signal

D. Narrow band FM signal

Q4.94.3 NAT 1994 1M

A 10 MHz carrier is frequency modulated by a sinusoidal signal of 500Hz, the maximum frequency deviation being 50 kHz. The bandwidth required (in kHz) as given by the Carson's rule is

Q4.93.1 MCQ 1993 2M

Which of the following demodulator(s) can be used for demodulating the signal

$$x(t) = 5(1 + 2\cos 200\pi t)\cos 20000\pi t$$

A. Envelope demodulator

B. Square-law demodulator

C. Synchronous demodulator

D. None of the above

Q4.93.2 NAT 1993 2M

A super-heterodyne radio receiver with an intermediate frequency of 455 KHz is tuned to a station operating at 1200 KHz. The associated image frequency (in kHz) is

Q4.92.1 MCQ 1992 2M

The maximum power efficiency of an AM modulator is

A. 25% **B.** 50%

C. 75% **D.** 100%

Q4.90.1 MCQ 1990 2M

In commercial TV transmission in India, picture and speech signals are modulated respectively as:

A. VSB and VSB **B.** VSB and SSB

C. VSB and FM **D.** FM and VSB

Q4.89.1 MCQ 1989 2M

A signal $x(t) = 2\cos(\pi \cdot 10^4 t)$ volts is applied to an FM modulator with the sensitivity constant of 10 kHz/volt. Then the modulation index of the FM wave is

A. 4 **B.** 2

C. $\frac{4}{\pi}$ **D.** $\frac{2}{\pi}$

Q4.89.2 MCQ 1989 2M

Which of the following schemes suffer(s) from the threshold effect?

A. AM detection using envelope detection

B. AM detection using synchronous detection

C. FM detection using a discriminator

D. SSB detection with synchronous detection

Q4.87.1 MCQ 1987 2M

A carrier $A_c \cos \omega_c t$ is frequency modulated by a signal $E_m \cos \omega_m t$. The modulation index is m_f. The expression for the resulting FM signal is:

A. $A_c \cos\left[\omega_c t + m_f \sin \omega_m t\right]$

B. $A_c \cos\left[\omega_c t + m_f \cos \omega_m t\right]$

C. $A_c \cos\left[\omega_c t + 2\pi m_f \cos \omega_m t\right]$

D. $A_c \cos\left[\omega_c t + \frac{2\pi m_f}{\omega_m} E_m \cos \omega_m t\right]$

Q4.87.2 MCQ 1987 2M

In a radar receiver the antenna is connected to the receiver through a waveguide. Placing the preamplifier on the antenna side of the waveguide rather than on the receiver side leads to:

A. A reduction in the overall noise figure.

B. A reduction in interference.

C. An improvement in selectivity characteristics.

D. An improvement in directional characteristics.

Answer Key

Q4.26.1 A, B, C	Q4.25.1 A,D	Q4.25.2 A
Q4.24.1 2.49 to 2.51	Q4.23.1 B	Q4.22.1 A
Q4.21.1 1400	Q4.21.2 1011310 to 1011320	Q4.21.3 94.2 to 94.6
Q4.20.1 2	Q4.20.2 B	Q4.19.1 0.74 to 0.76
Q4.18.1 D	Q4.18.2 0.12 to 0.13	Q4.17.1 70
Q4.17.2 5.19 to 5.23	Q4.16.1 4.9 to 5.1	Q4.16.2 3485

Q4.16.3 0.49 to 0.51	Q4.16.4 A	Q4.15.1 C
Q4.15.2 349 to 351	Q4.15.3 C	Q4.15.4 C
Q4.14.1 59.9 to 60.1	Q4.14.2 0.45 to 0.55	Q4.14.3 A
Q4.14.4 3.94 to 4.05	Q4.12.1 B	Q4.11.1 B
Q4.11.2 B	Q4.10.1 B	Q4.10.2 C
Q4.09.1 C	Q4.09.2 C	Q4.08.1 C

Q4.08.2 B	Q4.08.3 A	Q4.07.1 A
Q4.06.1 B	Q4.06.2 C	Q4.06.3 A
Q4.06.4 A	Q4.06.5 B	Q4.05.1 D
Q4.05.2 A	Q4.05.3 C	Q4.05.4 D
Q4.04.1 B	Q4.04.2 D	Q4.04.3 C
Q4.04.4 B	Q4.04.5 B	Q4.03.1 A

Q4.03.2 B	Q4.03.3 D	Q4.03.4 D
Q4.03.5 C	Q4.03.6 A	Q4.02.1 C
Q4.02.2 D	Q4.02.3 C	Q4.00.1 A
Q4.00.2 D	Q4.00.3 C	Q4.99.1 D
Q4.98.1 B	Q4.98.2 D	Q4.96.1 B
Q4.96.2 D	Q4.95.1 B	Q4.95.2 C
Q4.95.3 B	Q4.94.1 A-3,B-1,C-2	Q4.94.2 D
Q4.94.3 101	Q4.93.1 C	Q4.93.2 2110
Q4.92.1 B	Q4.90.1 C	Q4.89.1 A
Q4.89.2 C	Q4.87.1 A	Q4.87.2 A

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